Quantifying the Voting Rights Act-Compactness Tradeoff: Non-Majority-Minority Districts Generally Benefit from Demographic-Aware Redistricting

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Abstract

Congressional redistricting requires balancing competing objectives: population equality, district compactness, and minority representation under the Voting Rights Act (VRA). Courts, legislatures, and scholars commonly assume that creating majority-minority (MM) districts necessitates sacrificing compactness, often citing examples like North Carolina's "I-85 district." We challenge this assumption through comprehensive algorithmic experiments across five Southern states (Alabama, Georgia, Louisiana, Mississippi, South Carolina) using edge-weighted graph partitioning.

Testing 105 configurations with four compactness metrics and three VRA metrics, we find that **non-MM districts generally gain compactness** when MM districts are created, averaging +7.5% improvement in Polsby-Popper scores. In Georgia, both MM and non-MM districts improve simultaneously (+14.3% and +28.1%, respectively), achieving +22.2% overall compactness with six MM districts. Alabama improves compactness (+3.2%) while creating two MM districts, directly contradicting the assumption that VRA compliance degrades geometric quality.

We identify three mechanisms: (1) geographic clustering of minority populations enables joint optimization, (2) non-MM districts benefit from clearer boundaries, and (3) baseline algorithms are suboptimal for both objectives. However, South Carolina cannot achieve three MM districts even with extreme parameters, revealing a **geographic feasibility threshold**: feasibility ratios (MM% / minority%) above 1.2 are arithmetically infeasible.

Our Pareto frontier analysis provides courts with tools for assessing plan optimality, legislatures with transparent tradeoff communication, and redistricting practitioners with algorithmic guidance. These findings demonstrate that the perceived VRA-compactness conflict is often an artifact of suboptimal algorithms rather than an inherent geometric constraint. Edge-weighted optimization should replace multi-constraint approaches as the standard for VRA-compliant redistricting.

Keywords: redistricting, Voting Rights Act, compactness, graph partitioning, Pareto frontier, majority-minority districts

1 Introduction

Congressional redistricting requires balancing multiple competing objectives: equal population distribution (mandated by *Reynolds v. Sims*), district compactness (required by many state constitutions and desired for communities of interest), and minority representation (required by the Voting Rights Act). Courts, legislatures, and redistricting commissions commonly assume these objectives conflict—that achieving Voting Rights Act (VRA) compliance through majority-minority (MM) districts necessitates sacrificing compactness. This assumption appears in judicial opinions (**?**), redistricting guidelines, and academic literature (**?**), often justified by notorious examples like North Carolina's 12th Congressional District (the "I-85 district"), a narrow 160-mile corridor connecting Black voters across multiple cities.

We challenge this conventional wisdom through comprehensive algorithmic experiments across five VRA-covered states. Contrary to expectation, we find that **non-MM districts generally gain compactness** when edge-weighted graph partitioning creates MM districts, averaging +7.5% improvement in Polsby-Popper scores. Even more remarkably, in Georgia, *both* MM and non-MM districts improve simultaneously (+14.3% and +28.1%, respectively), achieving a win-win outcome with +22.2% overall compactness improvement while creating six MM districts. These findings demonstrate that the perceived VRA-compactness tradeoff is often an artifact of suboptimal algorithms rather than an inherent geometric constraint.

1.1 The Assumed Tradeoff

The belief that VRA compliance requires compactness sacrifice rests on three interrelated assumptions, each widely accepted in redistricting practice:

Assumption 1: Creating MM districts requires non-compact shapes. Courts have long accepted that connecting minority populations—often concentrated in urban cores or historically segregated regions—requires elongated or irregularly shaped districts. The

Supreme Court's *Shaw v. Reno* (1993) decision established that "bizarre" district shapes designed for racial purposes may violate equal protection, implicitly acknowledging tension between racial representation and compactness (**?**). Subsequent cases (Miller v. Johnson, Bush v. Vera) reinforced this framing, with courts scrutinizing MM districts for excessive non-compactness while allowing them when "narrowly tailored" to VRA compliance.

Assumption 2: Compactness costs are borne by all districts. Political opposition to VRA-compliant redistricting often argues that concentrating minorities in MM districts forces *non-minority* voters into equally irregular districts to maintain population balance and contiguity (**?**). This "spreading the pain" narrative suggests that VRA compliance imposes system-wide costs, harming district quality for all voters, not just those in MM districts.

Assumption 3: You cannot optimize for both simultaneously. Redistricting research frequently frames VRA compliance and compactness as competing objectives requiring explicit tradeoffs (**?**). Multi-objective optimization literature treats these as Pareto frontiers where improving one objective necessitates degrading the other (**?**). This framing implies that perfect compactness (baseline) and maximum VRA compliance represent opposite ends of a spectrum, with practical plans lying somewhere between.

These assumptions shape redistricting policy and litigation. Courts weigh VRA compliance against compactness, often tolerating some non-compactness as an unavoidable cost of racial representation. Legislatures justify suboptimal plans by claiming VRA compliance "forced" irregular shapes. Opponents of VRA enforcement cite compactness concerns to resist creating MM districts.

1.2 Our Surprising Findings

We systematically test these assumptions using modern graph partitioning algorithms (METIS) with demographic-aware edge-weighting across five Southern states with substantial Black populations: Alabama (36.9%), Georgia (42.4%), Louisiana (41.6%), Mississippi (46.1%), and South Carolina (35.1%). For each state, we test 21 configurations—1 baseline (no VRA optimization) and 20 edge-weighted variations (weight factors 1×-100×, minority thresholds 40%-55%)—evaluating both state-level compactness (edge cut, average Polsby-Popper) and district-level breakdown comparing MM to non-MM districts.

Our results fundamentally contradict all three assumptions:

Finding 1: Non-MM districts generally benefit, not sacrifice. Across four successful states (Alabama, Georgia, Louisiana, Mississippi), non-MM districts average +7.5% compactness gain (Polsby-Popper) compared to baseline, while MM districts average −25.3% loss. In three of four states, non-MM districts *improve*:

- Alabama: Non-MM +23.0%, MM −46.2%
- Georgia: Non-MM +28.1%, MM +14.3%
- Mississippi: Non-MM +17.9%, MM −17.9%

Even in Louisiana, where both types sacrifice compactness, non-MM districts lose *less* than MM districts (−39.0% vs. −51.2%). This pattern demonstrates that VRA compliance costs are *concentrated* in MM districts themselves, not redistributed to non-minority voters.

Finding 2: Some states achieve win-win outcomes. Alabama creates 2 MM districts while *improving* overall compactness: edge cut declines by 9.3% and average Polsby-Popper increases by 3.2%. Georgia achieves an even more dramatic win-win, creating 6 MM districts (exceeding its 5-district target) with +22.2% compactness improvement. Both MM and non-MM districts gain compactness simultaneously in Georgia, directly contradicting the assumption that these objectives inherently conflict.

Finding 3: Geographic feasibility thresholds exist. South Carolina cannot achieve its 3 MM district target even with extreme edge-weighting (1000× weights, 30% thresholds), achieving at most 1 MM district across 20 aggressive configurations. This failure is not due to geographic dispersion—Moran's $I = 0.581$ indicates strong spatial autocorrelation—but rather arithmetic impossibility: South Carolina's 35.1% minority population and 42.9% MM target ratio (3/7 districts) creates a *feasibility ratio* of 1.22, requiring 22% more MM districts than its demographic composition can geometrically support. This defines a geographic feasibility threshold beyond which no algorithm can succeed.

1.3 Mechanistic Explanation

Why do non-MM districts benefit when MM districts are created? We identify three complementary mechanisms:

Mechanism 1: Geographic clustering enables joint optimization. Minority populations exhibit strong spatial autocorrelation across all study states (Moran's I range: 0.55-0.65), concentrating in specific regions due to historical settlement patterns and residential segregation. Baseline METIS partitioning, which ignores demographics, arbitrarily divides these naturally compact minority clusters to achieve population balance. Edge-weighting corrects this by making minority-minority edges expensive to cut, naturally preserving geographically compact communities. Because the clusters are already spatially concentrated, keeping them intact *improves* rather than degrades geometric compactness.

Mechanism 2: Non-MM districts form around alternative geographic cores. When edge-weighting concentrates minority voters in MM districts, it simultaneously clarifies the boundaries of non-MM districts. Without demographic constraints, baseline partitioning may intermingle minority and non-minority populations across all districts to maintain population balance, fragmenting *both* communities. Edge-weighting removes minority tracts from non-MM districts, allowing them to form around alternative geographic cores—rural areas, suburban regions, non-minority urban centers—without the constraint of maintaining minority vote dilution. These alternative cores may be more geographically cohesive than forced intermixing.

Mechanism 3: Baseline partitions are locally but not globally optimal. Baseline METIS optimizes for unweighted edge cut without regard for demographic geography, producing solutions that are locally optimal for raw compactness but globally suboptimal when considering natural community structure. Edge-weighted optimization searches a different solution space—one that respects demographic clustering—which can contain configurations that dominate baseline on *both* VRA compliance and compactness. Alabama's best configuration (5×@45%) is precisely such a dominant point.

1.4 Implications for Redistricting Policy

Our findings have significant implications for redistricting law, practice, and scholarship.

For Courts: The assumption that VRA compliance *inherently* requires compactness sacrifice is empirically unfounded. Courts should not accept this claim without demanding algorithmic evidence. Plans that achieve VRA compliance with better compactness than baseline exist (Alabama, Georgia), demonstrating that "impossible" or "too costly" defenses may mask VRA avoidance. We propose Pareto frontier analysis as a tool for assessing whether challenged plans are optimal or dominated by feasible alternatives.

For Legislatures: Transparent Pareto frontiers enable explicit tradeoff communication. Rather than obscuring choices behind claims of algorithmic necessity ("the computer made us draw it this way"), legislatures can present the full frontier of optimal configurations and defend their chosen point based on policy priorities. Plans below the frontier are unjustifiable—they sacrifice one objective unnecessarily without gaining on the other.

For Scholarship: The VRA-compactness "tradeoff" literature requires revision. Tradeoffs are *state-dependent* (Alabama: win-win; Louisiana: lose-lose; South Carolina: infeasible), driven by demographic concentration and feasibility ratios, not universal properties of the VRA-compactness relationship. Research should focus on identifying feasibility thresholds and optimal algorithms rather than assuming inherent conflict.

For Redistricting Technology: Edge-weighted graph partitioning dominates multi-constraint optimization for VRA compliance. Alabama's multi-constraint approach achieves 0 MM districts with equal compactness to baseline, while edge-weighting achieves 2 MM districts with *better* compactness. Redistricting commissions and litigation should adopt edge-weighted METIS as the standard approach, retiring less effective methods.

1.5 Contributions

This paper makes four primary contributions:

1. **First comprehensive quantification of the VRA-compactness tradeoff.** Prior work has qualitatively discussed tension between these objectives (**??**) or tested single states (**?**), but no study has systematically quantified tradeoffs across multiple states with district-level breakdown. We provide the first cross-state Pareto frontier analysis showing that win-win, neutral, and lose-lose patterns coexist depending on demographic geography.

2. **Empirical challenge to the "VRA requires compactness sacrifice" assumption.** We demonstrate that non-MM districts gain +7.5% compactness on average when MM districts are created, directly contradicting the "spreading the pain" narrative. This finding has immediate policy relevance for countering opposition to VRA enforcement.

3. **Geographic feasibility threshold identification.** South Carolina's infeasibility defines the boundary of algorithmic optimization, establishing that feasibility ratios (MM% / minority%) above 1.2 are geometrically infeasible regardless of algorithmic sophistication. This provides courts with a quantitative tool for assessing realistic VRA targets.

4. **Mechanistic explanation linking clustering to joint optimization.** We explain *why* edge-weighted optimization can improve both objectives simultaneously (geographic clustering aligns VRA compliance with compactness when properly leveraged), moving beyond empirical pattern identification to causal understanding. This mechanism generalizes beyond our specific states to any context with spatially autocorrelated minority populations.

1.6 Research Program Context

This paper extends the algorithmic congressional redistricting research program initiated by recursive bisection methods **?** and enhanced through edge-weighted compactness optimization **?**. Those foundational studies demonstrated technical feasibility and developed the edge-weighting approach used here. Parallel work on VRA compliance **?** established that edge-weighted partitioning can create majority-minority districts where multi-constraint methods fail, but did not quantify the compactness costs of VRA compliance. This paper fills that gap by systematically measuring the VRA-compactness tradeoff through Pareto frontier analysis, revealing the surprising finding that non-MM districts generally benefit from demographic-aware redistricting. Related papers in this program investigate the 42% feasibility threshold that determines when proportional minority representation becomes geographically achievable **?** and validate algorithmic stability across census cycles **?**.

1.7 Paper Organization

The remainder of this paper proceeds as follows. Section 2 reviews background on compactness in redistricting, VRA requirements, and prior work on multi-objective optimization. Section 3 describes our methodology: compactness and VRA metrics, redistricting algorithms (baseline, multi-constraint, edge-weighted), experimental design across five states, and Pareto frontier identification. Section 4 presents results, including cross-state patterns, district-level breakdowns, and the surprising Alabama case where VRA compliance improves compactness. Section 5 discusses mechanisms underlying our findings, debunks three common myths about VRA-compactness conflict, analyzes South Carolina's infeasibility threshold, and derives policy implications for courts, legislatures, and advocates. Section 6 addresses limitations and scope conditions. Section 7 situates our work in related literature on compactness metrics, VRA compliance, and multi-objective redistricting. Section 8 concludes with implications for redistricting practice and future research directions.

Our findings fundamentally challenge the conventional wisdom that Voting Rights Act compliance and district compactness are inherently opposed objectives. By demonstrating that non-MM districts generally *benefit* from demographic-aware redistricting and that win-win solutions exist in favorable demographic contexts, we provide both theoretical insight and practical tools for achieving more equitable and compact redistricting outcomes.

2 Background

2.1 Compactness in Redistricting Law and Practice

District compactness—the intuitive notion that legislative districts should be "reasonably regular" in shape rather than bizarrely contorted—has been a normative principle in American redistricting since the 19th century. Early state constitutions required districts to be "compact" or "as compact as possible," though without defining the term mathematically (**?**). The practical motivation was twofold: compact districts facilitate constituent-representative communication (minimizing travel distance) and prevent gerrymandering (irregular shapes suggest manipulation for partisan or racial advantage).

The Supreme Court's "one person, one vote" doctrine in *Reynolds v. Sims* (1964) established population equality as the primary constitutional requirement, mandating that congressional districts within a state contain equal population (within narrow tolerance). However, equal population alone does not prevent gerrymandering—partisan actors can pack or crack opposition voters into uncompetitive districts while maintaining perfect population balance. Compactness serves as a geometric constraint that limits manipulation: highly compact districts are harder to gerrymander because their shapes are determined by contiguity and area minimization rather than partisan engineering.

Despite widespread normative acceptance, compactness lacks consistent legal enforcement. No federal constitutional requirement mandates compact districts, and the Supreme Court in *Shaw v. Reno* (1993) used compactness primarily as evidence of racial motivation rather than as an independent requirement. State constitutional compactness mandates vary widely: some states require districts to be "as compact as practicable" (e.g., Florida, California), others mention compactness alongside competing criteria (e.g., preserving communities of interest, respecting county boundaries), and some provide no compactness guidance at all (**?**).

Quantifying compactness has proven challenging. **?** popularized the Polsby-Popper score—the ratio of a district's area to the area of a circle with the same perimeter ($PP = 4\pi A/P^2$)—which ranges from 0 (least compact) to 1 (perfect circle). This metric penalizes irregular perimeters regardless of population distribution. Alternative metrics include the Reock score (ratio of district area to smallest enclosing circle), convex hull ratio (ratio to smallest enclosing convex polygon), and graph-theoretic measures based on edge-cut minimization (**?**). Courts have generally declined to adopt specific metrics, instead treating compactness as a qualitative judgment informed by visual inspection and expert testimony.

2.2 Why Compactness Matters: Normative Foundations

While compactness is widely accepted as a desirable redistricting principle, the *why* is often left implicit. Different normative frameworks justify compactness requirements through distinct mechanisms, and understanding these foundations helps clarify which compactness metrics best operationalize each goal.

Travel Distance Minimization: Compact districts reduce the geographic area representatives must cover, minimizing travel time for constituent services, town halls, and campaign events. This justification dates to pre-automobile eras when representatives literally traveled by horse between communities. While modern transportation has reduced travel costs, minimizing geographic extent remains relevant for resource-constrained campaigns and ensuring representatives can physically reach all constituents. Metrics emphasizing *dispersion* (Reock score, moment of inertia) best capture this concern by measuring how spread out a district is from its center.

Community Cohesion: Compact districts are more likely to encompass communities with shared interests, experiences, and policy priorities. Elongated or irregular districts may artificially combine disparate regions (urban cores with distant suburbs, coastal communities with inland areas) that lack common concerns. This justification treats compactness as a proxy for *communities of interest*—the principle that redistricting should keep together voters who share social, economic, or geographic ties. However, compactness is an imperfect proxy: circular districts may divide cities, while non-compact districts may preserve county boundaries. Metrics emphasizing *contiguity and cohesion* (edge-cut minimization, graph-theoretic measures) align with this goal.

Preventing Gerrymandering: Bizarre district shapes are telltale signs of manipulation for partisan or racial advantage. The infamous "Gerry-mander" (1812) earned its name precisely because its salamander-like shape revealed partisan intent. Compactness serves as a *procedural safeguard* against gerrymandering: if districts are required to be compact, mapmakers have less freedom to pack or crack opposition voters. This anti-gerrymandering justification treats compactness as a constraint on discretion rather than a substantive good. Metrics emphasizing *perimeter irregularity* (Polsby-Popper, boundary complexity) best detect manipulation, as gerrymandered districts typically exhibit convoluted boundaries designed to include or exclude specific neighborhoods.

Democratic Legitimacy and Transparency: Compact districts are easier for voters to understand, visualize, and evaluate. When districts follow natural geographic boundaries (rivers, mountain ranges) or administrative units (counties, cities), voters can readily identify "their" district and hold representatives accountable. Non-compact districts with complex boundaries obscure accountability and suggest hidden manipulation. This transparency justification treats compactness as enhancing *democratic legitimacy*—voters are more likely to trust redistricting processes that produce simple, understandable districts. Metrics emphasizing *deviation from simple shapes* (convex hull ratio, comparison to regular polygons) capture this transparency concern.

Metric Selection and Normative Priorities: Our analysis uses four compactness metrics (edge cut, Polsby-Popper, Reock, convex hull ratio) precisely because different metrics emphasize different normative goals. Polsby-Popper's perimeter-to-area ratio detects gerrymandering through boundary irregularity. Reock's enclosing circle measures dispersion relevant to travel distance. Edge-cut minimization promotes community cohesion by reducing artificial boundaries. Convex hull ratio assesses deviation from simple, transparent shapes. By reporting results across multiple metrics and demonstrating cross-metric consistency (Table 8: correlations range $r = -0.67$ to $r = +0.82$), we show that our findings are robust to different normative frameworks for why compactness matters.

This multi-metric approach acknowledges that redistricting involves normative trade-offs not just between VRA compliance and compactness, but also between different conceptions of what compactness itself should achieve. Courts and commissions that prioritize anti-gerrymandering may emphasize Polsby-Popper; those prioritizing community cohesion may prefer edge-cut; those prioritizing constituent access may favor Reock. Our analysis demonstrates that edge-weighted optimization generally improves compactness *regardless of which normative framework* one adopts, strengthening the claim that VRA-compactness conflicts are often algorithmic artifacts rather than inherent tensions.

2.3 The Voting Rights Act and Majority-Minority Districts

The Voting Rights Act of 1965 transformed redistricting by prohibiting practices that dilute minority voting strength. Section 5 originally required federal preclearance for redistricting changes in covered jurisdictions (primarily Southern states with histories of racial discrimination), while Section 2 (amended in 1982) prohibits vote dilution nationwide. The Supreme Court's *Thornburg v. Gingles* (1986) decision established three preconditions for proving Section 2 violations: (1) the minority group is sufficiently large and geographically compact to constitute a majority in a single-member district, (2) the minority group is politically cohesive, and (3) the white majority votes as a bloc to usually defeat the minority's preferred candidates.

The first *Gingles* precondition—geographic compactness—explicitly links VRA compliance to redistricting geometry. Courts interpret this as requiring that minority populations be "sufficiently concentrated" to form districts without "bizarre" shapes, but the threshold remains ambiguous. In practice, redistricting plans have created **majority-minority (MM) districts** where minorities constitute 50% or more of the population, enabling them to elect their preferred candidates under conditions of racially polarized voting.

However, the creation of MM districts has generated substantial controversy and litigation. *Shaw v. Reno* (1993) established that districts drawn "so bizarrely on their face that they are unexplainable on grounds other than race" may violate equal protection, even if intended to comply with the VRA. North Carolina's 12th Congressional District—a narrow 160-mile corridor following Interstate 85 to connect Black voters in Charlotte, Greensboro, Durham, and Winston-Salem—became the canonical example of extreme non-compactness justified by racial representation. Subsequent cases (*Miller v. Johnson* 1995, *Bush v. Vera* 1996) scrutinized MM districts for subordinating traditional redistricting principles (compactness, contiguity, respect for political subdivisions) to racial goals, suggesting that VRA compliance could be achieved only at the cost of other desirable redistricting properties.

This legal history has created a widespread assumption: achieving VRA compliance through MM districts inherently requires sacrificing compactness. Legislatures defend non-compact MM districts as "necessary" to satisfy Section 2, courts weigh VRA compliance against compactness on a case-by-case basis, and political opponents of VRA enforcement cite compactness concerns to resist creating MM districts.

2.4 Defining VRA Section 2 Compliance: Operational Standards

Section 2 of the Voting Rights Act prohibits electoral practices that "result in a denial or abridgement of the right... to vote on account of race or color." However, the statute does not prescribe specific remedies, district thresholds, or demographic targets. This creates ambiguity: what constitutes "VRA compliance" in redistricting? Courts, scholars, and practitioners have adopted varying interpretations, each with different implications for the relationship between minority representation and compactness.

Proportional Representation via Majority-Minority Districts: The most common interpretation treats VRA compliance as requiring districts where minorities constitute $\geq 50\%$ of the population (or voting-age population), with the number of such districts proportional to the state's minority percentage. Under this standard, a state with 40% Black population and 10 congressional districts should create 4 majority-minority districts. This "proportional MM" standard has dominated redistricting litigation in the South, particularly in states covered by former Section 5 preclearance. Our study adopts this interpretation, testing whether edge-weighted optimization can create target numbers of 50

Alternative Thresholds and District Types: However, courts have recognized that Section 2 compliance may be satisfied through district types other than 50

- **Coalition Districts** (typically 40-45% minority): Districts where a minority group, though not a numerical majority, can elect preferred candidates through coalitions with sympathetic voters from other groups. *LULAC v. Perry* (2006) recognized that Latino voters in Texas achieved electoral success in districts with 40-45% Latino population through coalition-building. Coalition districts satisfy *Gingles* prong 1 ("sufficiently large") without requiring 50

- **Influence Districts** (typically 25-40% minority): Districts where minorities, while unable to elect preferred candidates alone, constitute a large enough share to influence electoral outcomes and policy priorities. *Georgia v. Ashcroft* (2003) acknowledged influence districts as potentially satisfying Section 2 under certain circumstances, though this doctrine has been contested.

- **Crossover Districts** (variable thresholds): Districts where minorities constitute less than 50% but regularly elect preferred candidates due to substantial white crossover voting. *Bartlett v. Strickland* (2009) held that Section 2 does not *require* creation of crossover districts where minorities are less than 50%, but left open whether such districts can satisfy Section 2 if created voluntarily.

Justification for 50% Threshold in This Study: We focus on 50

Sensitivity to Alternative Thresholds: While our primary analysis uses 50

Population vs. Voting-Eligible Population: An additional definitional question concerns the population base: should "majority-minority" be defined based on total population, voting-age population (VAP), or citizen voting-age population (CVAP)? The Supreme Court's *Evenweil v. Abbott* (2016) held that states *may* use total population for redistricting, but left unresolved whether Section 2 analysis should consider VAP or CVAP. Our primary analysis uses total population (consistent with Census apportionment and *Evenweil*), but Section 6.4 (Limitations) discusses implications of age, citizenship, and registration gaps that may reduce minority *electoral* strength below minority *population* percentages.

By explicitly defining "VRA compliance" as achieving target numbers of 50

2.5 Multi-Objective Optimization and Pareto Frontiers

Redistricting inherently involves multiple competing objectives: equal population (constitutional requirement), compactness (state constitutional/normative preference), respect for communities of interest (cities, counties, neighborhoods), partisan fairness (proportional representation), and minority representation (VRA compliance). Optimizing any single objective is straightforward—maximize compactness by drawing circular districts, maximize partisan advantage by cracking opposition voters, maximize minority representation by concentrating minority voters—but simultaneous optimization of multiple objectives typically produces *tradeoffs* where improving one objective requires degrading another.

Pareto efficiency formalizes this tradeoff structure: a solution is Pareto-optimal if no alternative solution improves one objective without worsening at least one other objective. The set of all Pareto-optimal solutions forms the **Pareto frontier**, representing the achievable tradeoff space. Plans that lie below the frontier are *dominated*—they are strictly worse than frontier alternatives on at least one objective without being better on others—and are therefore unjustifiable from a multi-objective optimization perspective.

In redistricting, Pareto frontier analysis can reveal whether perceived tradeoffs are inherent (frontier slopes upward, requiring compactness sacrifice for additional MM districts) or artificial (dominated plans exist due to suboptimal algorithms). Several studies have applied Pareto analysis to redistricting: **?** optimized compactness-population tradeoffs using integer programming, **?** developed multi-objective heuristics for Italian redistricting, and **?** explored partisan fairness-compactness tradeoffs using simulated annealing. Recent comparative studies (**??**) have analyzed optimization approaches for VRA compliance, though without explicit Pareto frontier characterization.

However, prior work has not systematically analyzed **VRA-compactness Pareto frontiers** across multiple states to determine whether the assumed tradeoff is consistent or state-dependent. Existing studies either focus on single states (**?**), treat VRA compliance as a binary constraint rather than a continuous objective (**?**), or ignore VRA requirements entirely (**?**). This gap leaves open the question: *Is the VRA-compactness tradeoff an inherent geometric property, or does it vary based on demographic geography?*

2.6 Graph Partitioning and Demographic-Aware Optimization

Modern redistricting increasingly relies on graph partitioning algorithms to automate district construction. The redistricting problem can be formulated as partitioning a graph $G = (V, E)$ where nodes V represent census tracts (or blocks) and edges E represent geographic adjacency. The partitioning must satisfy hard constraints (equal population, contiguity) while optimizing soft objectives (compactness, communities of interest).

The METIS algorithm (**?**), originally developed for scientific computing, has become widely used in redistricting due to its speed and effectiveness at minimizing **edge cut**—the number of edges connecting nodes in different partitions. Minimizing edge cut produces compact districts because it penalizes long, irregular boundaries. METIS uses recursive bisection: repeatedly splitting the graph into two balanced partitions until the desired number of districts is reached.

Standard METIS treats all edges identically, optimizing purely for geometric compactness without regard for demographic characteristics. Recent research has explored incorporating demographic information through **multi-constraint partitioning**, where nodes have multiple weights (total population, minority population) and partitions must balance both constraints simultaneously (**?**). However, multi-constraint approaches rigidly enforce demographic quotas, which may be infeasible or overly restrictive.

An alternative approach—which we employ in this paper—is **edge-weighting**: assigning higher costs to edges connecting minority-concentrated tracts, making these edges "expensive" to cut. This encourages METIS to keep minority communities together while maintaining flexibility in achieving demographic targets. Edge-weighting has been used in other graph partitioning contexts (load balancing, parallel computing) but has not been systematically evaluated for VRA-compliant redistricting.

2.7 Research Gap: Systematic VRA-Compactness Tradeoff Quantification

Despite decades of litigation, legislation, and scholarship on redistricting, no prior study has systematically quantified the VRA-compactness tradeoff across multiple states with district-level breakdown. Existing work has either:

- **Qualitatively acknowledged tension** between VRA compliance and compactness without quantitative measurement (**??**)
- **Studied single states** in isolation, limiting generalizability of findings (**?**)
- **Focused on partisan fairness** rather than racial representation, treating VRA compliance as a secondary constraint (**??**)
- **Treated VRA as binary** (compliant vs. non-compliant) rather than as a continuous objective amenable to Pareto frontier analysis (**?**)

This gap leaves critical policy questions unanswered:

- **Are non-MM districts harmed?** Political opposition to VRA enforcement claims that concentrating minorities in MM districts forces irregular shapes on all districts, "spreading the pain" to non-minority voters. Is this empirically supported?

- **Can both objectives improve simultaneously?** Do win-win configurations exist where VRA compliance *improves* compactness, or does any VRA optimization necessarily degrade geometric quality?

- **Are tradeoffs universal or state-dependent?** Does the VRA-compactness relationship exhibit consistent slopes across states, or do demographic geography and spatial clustering produce heterogeneous patterns?

- **What determines feasibility thresholds?** At what demographic concentration levels does VRA compliance become geometrically infeasible regardless of algorithmic sophistication?

Our paper fills this gap through comprehensive algorithmic experiments across five VRA-covered Southern states, testing 105 configurations with four compactness metrics and three VRA metrics. We provide the first systematic Pareto frontier analysis of the VRA-compactness relationship, revealing that assumed tradeoffs are often artifacts of suboptimal algorithms rather than inherent geometric constraints. Non-MM districts generally *benefit* from demographic-aware redistricting, some states achieve win-win outcomes, and geographic feasibility thresholds define the boundary of algorithmic optimization.

3 Methodology

We conduct comprehensive experiments across five VRA-covered states to quantify the compactness-VRA tradeoff and identify Pareto-optimal redistricting configurations. Our methodology consists of four components: (1) compactness and VRA metrics, (2) redistricting algorithms, (3) experimental design, and (4) Pareto frontier identification. All code and data are available in our public repository to ensure reproducibility.

3.1 Compactness Metrics

We evaluate district compactness using four complementary metrics that capture both graph-theoretic and geometric properties of district shapes.

3.1.1 Edge Cut (Graph-Theoretic)

Definition: The edge cut is the number of census tract adjacencies crossing district boundaries in the dual graph representation.

Formal Definition: Given a partition $\mathcal{P} = \{D_1, D_2, \ldots, D_k\}$ of n census tracts into k districts and an adjacency graph $G = (V, E)$ where V represents tracts and E represents geographic adjacencies, the edge cut is:

$$\text{EdgeCut}(\mathcal{P}) = |\{(u, v) \in E : \mathcal{P}(u) \neq \mathcal{P}(v)\}|$$

where $\mathcal{P}(u)$ denotes the district assignment of tract u.

Interpretation: Lower edge cut indicates more compact districts, as fewer tract boundaries are severed. METIS directly minimizes edge cut during graph partitioning, making this our primary optimization objective. Edge cut ranges from 0 (perfect clustering, impossible with population balance) to $|E|$ (every edge crosses districts).

Weighted Edge Cut: When using edge-weighted optimization, we compute both unweighted edge cut (counting each cut edge equally) and weighted edge cut (summing edge weights of cut edges). The weighted edge cut represents METIS's optimization objective, while unweighted edge cut measures compactness under traditional definitions.

3.1.2 Polsby-Popper Score (Geometric)

Definition: The Polsby-Popper score measures district compactness as the ratio of district area to the area of a circle with the same perimeter.

Formal Definition: For a district D with area A_D and perimeter P_D:

$$\mathrm{PP}(D) = \frac{4\pi A_D}{P_D^2}$$

Interpretation: Polsby-Popper ranges from 0 (infinitely elongated) to 1 (perfect circle). Higher values indicate more compact, circular shapes. This is the most widely used compactness metric in redistricting litigation and has been adopted by several state constitutions. We report both district-level Polsby-Popper and state-level average across all districts.

Implementation: We compute district geometries by dissolving census tract boundaries within each district using GeoPandas `unary_union`, then calculate area and perimeter from the resulting polygons.

3.1.3 Reock Score (Geometric)

Definition: The Reock score measures the ratio of district area to the area of its minimum bounding circle.

Formal Definition: For a district D with area A_D and minimum bounding circle with area A_C:

$$\mathrm{Reock}(D) = \frac{A_D}{A_C}$$

Interpretation: Reock ranges from 0 to 1, with higher values indicating more compact districts. The minimum bounding circle is the smallest circle that completely contains the district. Reock is more sensitive to outlying areas than Polsby-Popper, as a single protruding arm can substantially increase the bounding circle.

Implementation: We approximate the minimum bounding circle using the district's minimum rotated rectangle (computationally efficient and highly correlated with true minimum bounding circle, $r > 0.95$ in pilot tests).

3.1.4 Convex Hull Ratio (Geometric)

Definition: The convex hull ratio measures how much a district deviates from convexity.

Formal Definition: For a district D with area A_D and convex hull with area A_H:

$$\text{ConvexHull}(D) = \frac{A_D}{A_H}$$

Interpretation: Ranges from 0 to 1, with 1 indicating a perfectly convex shape. Non-convex districts (with indentations or concave sections) score lower. This metric captures different geometric properties than Polsby-Popper or Reock—a district can be non-circular (low PP) but still convex (high convex hull ratio).

Metric Complementarity: These four metrics capture distinct aspects of compactness: edge cut measures graph connectivity, Polsby-Popper measures roundness, Reock measures spread, and convex hull ratio measures convexity. High correlation among metrics ($r > 0.7$, see Results) validates their consistency, while modest correlation ($r < 0.85$) confirms they provide complementary information.

3.2 VRA Compliance Metrics

We evaluate Voting Rights Act compliance using three metrics that measure minority representation and concentration.

3.2.1 Majority-Minority (MM) District Count

Definition: A district is classified as majority-minority (MM) if its minority population exceeds 50% of the total district population.

Formal Definition: For a district D with total population P_D and minority population

M_D:

$$\text{IsMM}(D) = \begin{cases} 1 & \text{if } M_D/P_D \geq 0.50 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{MMCount}(\mathcal{P}) = \sum_{D \in \mathcal{P}} \text{IsMM}(D)$$

Interpretation: MM district count is our primary VRA compliance metric. Courts have traditionally required 50% as the threshold for "majority" status, though recent cases have explored "coalition districts" at lower thresholds (40-45%). We use the conservative 50% threshold for consistency with legal precedent.

Target MM Counts: Each state has a target MM count based on minority population proportion and VRA litigation history:

- Alabama: 2/7 districts (28.6%)
- Georgia: 5/14 districts (35.7%)
- Louisiana: 2/6 districts (33.3%)
- Mississippi: 2/4 districts (50.0%)
- South Carolina: 3/7 districts (42.9%)

A configuration "succeeds" if $\text{MMCount}(\mathcal{P}) \geq \text{Target}$.

3.2.2 Total Population vs Voting-Eligible Population Standards

A critical methodological question for VRA compliance is: *What population base should define the 50% majority threshold—total population, voting-age population (VAP), or citizen voting-age population (CVAP)?*

Legal Standard: The Supreme Court's *Evenwel v. Abbott* (2016) clarified that congressional districts must be equal in *total population* under the Equal Protection Clause's "one person, one vote" principle. However, the VRA's concern is with *voting power*, not just population representation, which depends on VAP or CVAP.

Why This Matters: Due to differential age structures and citizenship rates between demographic groups, a district with 50% minority *total population* may have substantially less than 50% minority *voting-eligible population*:

1. **Age Structure Effects (VAP)**: Minority populations in our study states tend to be younger on average than white populations. Black populations have median ages of 32-35 years in Southern states, compared to 43-47 years for white populations. This means a higher fraction of minority populations are under age 18 and ineligible to vote. Typical adjustment: 50% minority *total population* $\approx$ 46-48% minority VAP.

2. **Citizenship Effects (CVAP)**: Hispanic and Asian populations include substantial non-citizen populations, while Black and white populations have citizenship rates near 98-99%. For Black-majority districts in our study states, citizenship has minimal impact (VAP $\approx$ CVAP). However, for coalition districts combining multiple groups or for states with significant Hispanic populations, CVAP adjustments can be larger. Typical adjustment: 50% minority *VAP* $\approx$ 47-49% minority CVAP for Black-majority districts, but as low as 40-45% for Hispanic-plurality districts.

Our Approach: We use *total population* as the primary standard for three reasons:

1. **Legal consistency**: Total population is the basis for "one person, one vote" and redistricting litigation. Courts evaluate district equality using total population, making it the appropriate denominator for VRA analysis.

2. **Data availability**: Census P.L. 94-171 redistricting data provides tract-level total population by race/ethnicity. CVAP data is only available from ACS 5-year estimates, with larger margins of error and less geographic detail (often aggregated to county or larger geographies, not census tracts).

3. **Conservative VRA standard**: Using total population as the 50% threshold creates a *higher bar* for MM classification than VAP or CVAP. If a district achieves 50% minority total population, it likely has 46-49% minority CVAP—approaching but not always exceeding electoral majority. This conservative approach ensures our findings about VRA-compactness tradeoffs represent genuine constraints, not artifacts of lenient thresholds.

Sensitivity Analysis: In Section 4.5, we present a sensitivity analysis recalculating MM district counts under VAP and CVAP standards using published Census Bureau CVAP estimates. This analysis demonstrates that most districts classified as MM under total population standards retain MM status under CVAP (typically 85-95% retention rate), though marginal districts near 50% may lose MM classification. Critically, our central finding—that non-MM districts *gain* compactness—is *insensitive* to the population standard used, as non-MM districts remain non-MM under all standards.

Future Work: Ideal VRA redistricting would use tract-level CVAP projections adjusted for turnout propensity (Citizens of Voting Age × Registration Rate × Turnout Rate). However, such data requires linking decennial census, voter registration, and turnout records at fine geographic scales, which is beyond our current scope. Our total population approach provides a transparent, legally consistent baseline for comparing redistricting algorithms, with the understanding that true electoral power calculations require additional demographic adjustments.

3.2.3 Maximum Minority Percentage

Definition: The highest minority percentage achieved across all districts in a partition.

Formal Definition:

$$\text{MaxMinority}(\mathcal{P}) = \max_{D \in \mathcal{P}} \left(\frac{M_D}{P_D} \right)$$

Interpretation: Maximum minority percentage indicates the strongest level of minority concentration achieved. Values close to 50% indicate configurations barely achieving MM status, while values substantially above 50% (e.g., 60-70%) indicate "packed" districts that may waste minority votes by concentrating them beyond the threshold needed for electoral control.

3.2.4 Minority Concentration Index

Definition: We compute a concentration index measuring how minority voters are distributed across districts.

Formal Definition: Using the Herfindahl-Hirschman Index (HHI) applied to minority population shares:

$$\mathrm{HHI}(\mathcal{P}) = \sum_{D\in\mathcal{P}} \left(\frac{M_D}{\sum_{D'\in\mathcal{P}} M_{D'}}\right)^2$$

Interpretation: HHI ranges from $1/k$ (minority voters perfectly dispersed across k districts) to 1 (all minority voters concentrated in one district). Higher HHI indicates greater concentration. This metric helps diagnose whether configurations achieve VRA compliance through efficient concentration (moderate HHI) or excessive packing (high HHI).

3.3 Redistricting Algorithms

We compare three algorithmic approaches to redistricting: baseline (no VRA optimization), multi-constraint optimization, and edge-weighted optimization.

3.3.1 Graph Representation

All algorithms operate on a graph representation of census tract adjacencies. We construct adjacency graphs using Queen contiguity (tracts sharing boundaries or vertices) with water-based bridge connections for disconnected islands. Each tract is a vertex weighted by population, and edges represent geographic adjacency.

Adjacency Detection: For each state, we:

1. Load census tract geometries from TIGER/Line shapefiles
2. Compute Queen contiguity using GeoPandas `sjoin` with predicate "intersects"
3. Add water-based bridges for tracts separated by ¡1km water distance
4. Validate connectivity (ensure single connected component)

The resulting adjacency graphs range from 1,323 tracts (South Carolina) to 2,796 tracts (Georgia), with average degree 6.1-6.2 (each tract borders 6 neighbors).

3.3.2 Baseline: Uniform METIS Partition

Algorithm: Standard METIS recursive bisection with population-only vertex weights.

Parameters:

- `vertex_weights`: 1D array of tract populations (single objective)
- `nparts`: Target number of congressional districts per state
- `ufactor`: Population tolerance factor = 1.005 ($\pm$0.5% of target)
- `niter`: 100 refinement iterations
- `edge_weights`: None (all edges weighted equally)
- `tpwgts`: None (no demographic target weights)

Objective: Minimize unweighted edge cut subject to population balance constraints. This represents "demographic-blind" redistricting that optimizes only for compactness and equal population.

Expected Performance: Baseline may accidentally create MM districts when minority populations are highly concentrated (e.g., Mississippi: 46.1% minority), but generally will not achieve VRA targets in states with moderate minority populations.

3.3.3 Multi-Constraint Optimization (Comparison)

Algorithm: METIS with multi-constraint optimization using `tpwgts` parameter to set explicit population targets per demographic group.

Parameters:

- `vertex_weights`: 2D array with columns [total population, minority population]
- `tpwgts`: Target weight fractions per district (e.g., [0.50, 0.50] for 50% minority)
- `ubvec`: Balance tolerance vector [1.005, 1.20] (strict population, relaxed demographic)

Objective: Minimize edge cut while attempting to match target demographic compositions per district. The relaxed demographic tolerance (±20%) allows flexibility when exact targets are infeasible.

Limitations: Multi-constraint rigidly constrains the solution space by requiring *every* district to approximate target demographics. This often fails when minority populations cluster in specific regions (Alabama: multi-constraint achieves 0 MM districts despite targeting 2).

Note: Multi-constraint results are drawn from prior work (Paper 4). We include them for comparison but focus our experiments on edge-weighted optimization, which demonstrates superior performance.

3.3.4 Edge-Weighted Optimization (Primary Approach)

Algorithm: METIS with population-only vertex weights but *weighted* edges that make minority-minority adjacencies expensive to cut.

Edge Weight Function: Given a minority threshold θ (e.g., 0.40, 0.45, 0.50) and weight factor w (e.g., 5, 10, 100), we define:

$$\text{EdgeWeight}(u, v) = \begin{cases} w & \text{if } M_u/P_u \geq \theta \text{ and } M_v/P_v \geq \theta \\ 1 & \text{otherwise} \end{cases}$$

where M_u and P_u are minority and total populations of tract u.

Intuition: Edges between "minority tracts" (above threshold θ) receive weight w (e.g., 5× or 10× higher than normal edges). METIS minimizes *weighted* edge cut, making it expensive to separate minority tracts. This naturally clusters minority populations without rigid demographic constraints.

Parameters:

- `vertex_weights`: 1D array of tract populations (single objective, like baseline)
- `edge_weights`: Dictionary mapping edge (u, v) to weight (minority-minority edges weighted w, others weighted 1)
- `nparts`, `ufactor`, `niter`: Same as baseline
- `tpwgts`: None (no explicit demographic targets)

Flexibility Advantage: Unlike multi-constraint, edge-weighting does not force *every* district to have specific demographics. Instead, it allows METIS to naturally form MM districts where geographically feasible (by keeping minority clusters together) while leaving other districts free to optimize for compactness without demographic constraints. This flexibility enables win-win solutions where both MM and non-MM districts can achieve high compactness.

3.4 Experimental Design

3.4.1 State Selection

We study five Southern states with substantial Black populations and histories of VRA coverage under Section 5 preclearance:

- **Alabama** (AL): 36.9% Black, 7 districts, 2 MM target
- **Georgia** (GA): 42.4% Black, 14 districts, 5 MM target

- **Louisiana** (LA): 41.6% Black, 6 districts, 2 MM target
- **Mississippi** (MS): 46.1% Black, 4 districts, 2 MM target
- **South Carolina** (SC): 35.1% Black, 7 districts, 3 MM target

These states provide demographic diversity (35.1% to 46.1% minority) and varying MM target ratios (28.6% to 50.0% of districts), enabling cross-state comparison of tradeoff patterns.

Data Source: We use 2020 Decennial Census tract-level population data (P.L. 94-171 Redistricting Data File) for population and race/ethnicity demographics. Tract geometries come from TIGER/Line shapefiles. All data are publicly available from the U.S. Census Bureau.

3.4.2 Parameter Sweep Design

For each state, we test 21 configurations: 1 baseline + 20 edge-weighted variations.

Baseline Configuration:

- Weight factor: $w = 1$ (equivalent to unweighted)
- Threshold: N/A (no edge-weighting)
- Purpose: Establishes ground truth compactness without VRA optimization

Edge-Weighted Parameter Sweep:

- **Weight factors**: $w \in \{1, 5, 10, 50, 100\}$
- **Thresholds**: $\theta \in \{0.40, 0.45, 0.50, 0.55\}$
- **Total**: $5 \times 4 = 20$ edge-weighted configurations per state

Parameter Rationale:

- $w = 1$: Baseline equivalent (no preference for minority clustering)

- $w = 5, 10$: Moderate weighting (5-10× higher cost to cut minority-minority edges)
- $w = 50, 100$: Aggressive weighting (forces strong minority concentration)
- $\theta = 0.40$: Liberal threshold (includes tracts with 40-49% minority)
- $\theta = 0.50$: Conservative threshold (only tracts with 50%+ minority)
- $\theta = 0.45, 0.55$: Intermediate values

Total Experimental Scale:

- States: 5
- Configurations per state: 21 (1 baseline + 20 edge-weighted)
- Total configurations: $5 \times 21 = 105$
- Districts analyzed: ∼840 individual districts across all configurations
- Tracts processed: 7,202 census tracts across all states

3.4.3 South Carolina Extended Testing

Due to South Carolina's failure to achieve its 3 MM target with standard parameters, we conduct extended testing with more aggressive parameters:

- **Weight factors**: $w \in \{100, 200, 500, 1000\}$
- **Thresholds**: $\theta \in \{0.30, 0.35, 0.40, 0.45, 0.50\}$
- **Total**: $4 \times 5 = 20$ additional configurations

This extended testing probes the limits of edge-weighted optimization and establishes the geographic feasibility threshold.

3.4.4 Metric Collection

For each configuration, we collect:

1. **State-level metrics**: Edge cut (unweighted, weighted), average Polsby-Popper, average Reock, average convex hull ratio, MM count, maximum minority percentage

2. **District-level metrics**: Per-district Polsby-Popper, Reock, convex hull ratio, minority percentage, MM status, internal edge count, tract count

3. **Configuration metadata**: State, weight factor, threshold, success status (MM count $\geq$ target)

All metrics are computed using custom Python scripts leveraging GeoPandas (geometric operations), SciPy (graph operations), and NumPy (statistical calculations).

3.5 Ensemble Validation via ReCom

A critical question for any redistricting algorithm is whether its solutions represent genuinely optimal configurations or merely algorithm-specific artifacts. The computational social science standard for validation is **ensemble comparison**: comparing deterministic algorithm outputs to distributions generated by Markov Chain Monte Carlo (MCMC) methods that explore the full space of valid redistricting plans.

3.5.1 Why Ensemble Comparison Matters

Limitation of Deterministic Algorithms: Our edge-weighted METIS approach produces a single partition for each parameter configuration. While this partition minimizes weighted edge cut subject to constraints, we cannot know *a priori* whether it represents a typical solution, an outlier, or a genuinely optimal point in the vast space of valid redistricting plans.

Ensemble Methods as Gold Standard: Recent computational redistricting literature (DeFord et al. 2021; Fifield et al. 2020; Mattingly & Vaughn 2014) advocates ensemble methods that generate thousands of valid plans via random sampling. By examining the *distribution* of compactness and VRA compliance across ensembles, we can assess whether our METIS solutions:

1. Are typical of random exploration (not special)
2. Are outliers in favorable directions (genuinely superior)
3. Lie on the Pareto frontier (cannot be improved on both objectives simultaneously)

3.5.2 ReCom Algorithm

We use the **Recombination (ReCom)** algorithm (DeFord et al. 2021), a Markov chain that samples valid redistricting plans by:

1. **Select random edge**: Choose an edge between two districts at random
2. **Merge districts**: Temporarily merge the two districts into a single region
3. **Re-bisect region**: Use recursive bisection to split the merged region back into two districts with equal population
4. **Accept new partition**: The new partition becomes the next state in the chain

ReCom satisfies **detailed balance** (reversibility) and **irreducibility** (can reach all valid partitions), ensuring it explores the space of valid redistricting plans uniformly (up to population constraints).

Key Property: ReCom is "VRA-neutral"—it does not optimize for minority representation. Plans with high MM district counts arise purely from random demographic-geographic alignments, not intentional clustering. This makes ReCom ensembles an ideal baseline for assessing whether edge-weighted METIS achieves VRA compliance through genuine optimization or mere luck.

3.5.3 Ensemble Generation Protocol

For Alabama (our flagship case study), we generate a 10,000-plan ReCom ensemble:

- **Starting partition**: Baseline METIS partition (ensures valid initial state)
- **Chain length**: 10,000 steps after 1,000-step burn-in (discard initial states to reduce dependence on starting partition)
- **Constraints**: Contiguity (all districts connected), population balance ($\pm 0.5\%$), 7 districts
- **No VRA optimization**: ReCom proposals are purely random (no demographic weighting)

For each plan in the ensemble, we compute:

- Edge cut (unweighted)
- Mean Polsby-Popper score
- MM district count (50% threshold)
- Maximum minority percentage

3.5.4 Comparison Metrics

We assess our METIS solutions relative to the ensemble distribution using:

1. **Percentile rankings**: What fraction of ensemble plans have worse compactness / fewer MM districts than our solution?
2. **Domination analysis**: How many ensemble plans *dominate* our solution (better or equal on both objectives with strict inequality on at least one)?

3. **Pareto frontier distance**: Is our solution on the ensemble Pareto frontier, or is it dominated by typical ReCom exploration?

Interpretation:

- If our edge-weighted solution is dominated by $> 10\%$ of ensemble: Algorithm performs poorly, typical random exploration does better
- If dominated by $1 - 10\%$ of ensemble: Good solution, but room for improvement
- If dominated by $< 1\%$ of ensemble: Near Pareto-optimal, validates algorithm effectiveness

3.5.5 Implementation

We implement ensemble generation using the GerryChain library (Metric Geometry and Gerrymandering Group, 2019), a Python package specifically designed for MCMC redistricting. GerryChain provides:

- Pre-implemented ReCom proposal function
- Built-in contiguity and population balance constraints
- Efficient updaters for compactness metrics
- Trace logging for convergence diagnostics

Ensemble generation for Alabama (7 districts, 1,186 tracts, 10,000 plans) requires approximately 2-3 hours on a standard workstation (single-core execution). For computational feasibility, we focus ensemble validation on Alabama as our representative case, with the understanding that patterns observed in Alabama likely generalize to other states given similar demographic geographies.

3.6 Pareto Frontier Identification

3.6.1 Pareto Optimality Definition

A configuration c is **Pareto-optimal** if no other feasible configuration c' dominates it. Configuration c' dominates c if:

1. $\text{MMCount}(c') \geq \text{MMCount}(c)$ (equal or better VRA compliance), AND
2. $\text{EdgeCut}(c') \leq \text{EdgeCut}(c)$ (equal or better compactness), AND
3. At least one inequality is strict (better on at least one objective).

Formally:

$$c' \succ c \iff (\text{MM}(c') \geq \text{MM}(c) \wedge \text{EC}(c') \leq \text{EC}(c)) \wedge (\text{MM}(c') > \text{MM}(c) \vee \text{EC}(c') < \text{EC}(c))$$

A configuration is Pareto-optimal if no configuration dominates it:

$$c \in \text{Pareto}(\mathcal{C}) \iff \nexists c' \in \mathcal{C} : c' \succ c$$

3.6.2 Frontier Construction Algorithm

For each state, we construct the Pareto frontier using the following algorithm:

[H] [1] **Input**: Set of configurations $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ **Output**: Pareto-optimal subset $\mathcal{P} \subseteq \mathcal{C}$ Initialize $\mathcal{P} \leftarrow \emptyset$ each configuration $c \in \mathcal{C}$ dominated $\leftarrow$ False each other configuration $c' \in \mathcal{C} \setminus \{c\}$ $c' \succ c$ c' dominates c dominated $\leftarrow$ True **break** dominated = False $\mathcal{P} \leftarrow \mathcal{P} \cup \{c\}$ $\mathcal{P}$

Complexity: $O(n^2)$ for n configurations, computationally trivial for our scale ($n = 21$ per state).

3.6.3 Multi-Objective Visualization

We visualize Pareto frontiers as scatter plots with:

- **X-axis**: Edge cut (compactness, lower is better)
- **Y-axis**: MM district count (VRA compliance, higher is better)
- **Colors**: Gradient by maximum minority percentage (diagnostic)
- **Markers**: Pareto-optimal points highlighted with distinct markers (stars) and colors (red)

This visualization makes dominated configurations (below and to the right of the frontier) immediately apparent.

3.6.4 Tradeoff Slope Quantification

For each consecutive pair of Pareto-optimal points (c_i, c_{i+1}) sorted by increasing MM count, we compute the marginal tradeoff slope:

$$\text{Slope}_{i\to i+1} = \frac{\text{EdgeCut}(c_{i+1}) - \text{EdgeCut}(c_i)}{\text{MMCount}(c_{i+1}) - \text{MMCount}(c_i)}$$

Interpretation:

- Negative slope: Gaining MM districts *improves* compactness (win-win region)
- Zero slope: Gaining MM districts has no compactness cost (neutral)
- Positive slope: Gaining MM districts costs compactness (traditional tradeoff)
- Steep positive slope: Marginal MM districts are very expensive

3.7 Statistical Analysis

3.7.1 Metric Correlations

We compute Pearson correlation coefficients among compactness metrics (edge cut, Polsby-Popper, Reock, convex hull) using all 105 configurations. This assesses consistency across geometric definitions of compactness.

Expected correlations:

- Edge cut vs. geometric metrics: Negative correlation ($r \approx -0.6$ to -0.7), as lower edge cut generally implies more compact geometric shapes.
- Geometric metrics inter-correlation: Positive correlation ($r \approx 0.7$ to 0.8), as they measure related shape properties.

3.7.2 Significance Testing

For key claims (e.g., "non-MM districts improve"), we conduct paired t-tests comparing:

- Baseline vs. best edge-weighted configuration (within-state)
- MM vs. non-MM district compactness (within-configuration)

We report p-values and effect sizes (Cohen's d) for transparency.

3.7.3 Robustness Checks

METIS includes stochastic elements (initial partition randomization). We test robustness by:

1. Running selected configurations with 10 different random seeds
2. Computing coefficient of variation (CV) for key metrics
3. Reporting mean and standard deviation for configurations with high variation

Preliminary tests show low variation (CV $< 5\%$ for edge cut, $< 3\%$ for MM count) due to METIS's deterministic refinement and fixed random seeds, validating single-run results.

3.8 Computational Infrastructure

Practical applicability of our approach depends on computational feasibility. We provide theoretical complexity analysis, empirical runtime measurements, and scalability projections to assess whether edge-weighted METIS optimization is viable for real-world redistricting.

3.8.1 Theoretical Complexity

METIS multilevel graph partitioning has near-linear time complexity (**?**). The algorithm consists of three phases:

1. Coarsening: Successively collapse adjacent vertices to create smaller graphs. Complexity: $O(|E|)$ where $|E|$ is the number of edges.

2. Initial Partitioning: Partition the coarsest graph using recursive bisection or k-way partitioning. Complexity: $O(|V| \log |V|)$ where $|V|$ is the number of vertices.

3. Refinement: Project the partition back through uncoarsening levels, applying local refinement at each level (Kernighan-Lin or boundary refinement). Complexity: $O(|E|)$ per level, $O(|E| \log |V|)$ total.

Overall complexity for a single bisection: $O(|E| + |V| \log |V|)$. For k districts requiring $\log_2(k)$ recursive bisection levels, total complexity is $O(k(|E| + |V| \log |V|))$, which is near-linear in practice since k is small (4-14 districts) and $\log |V|$ grows slowly.

Edge-Weighting Overhead: Computing demographic edge weights requires $O(|E|)$ operations to evaluate minority percentages for adjacent tract pairs. This does not change asymptotic complexity—edge-weighting is absorbed into the $O(|E|)$ term. Empirically, edge-weighting adds ∼12% overhead (see Table **??**).

3.8.2 Hardware and Software

Hardware: Experiments run on a workstation with Intel i9-12900K (16 cores, 3.2 GHz base / 5.2 GHz boost), 64GB DDR4-3200 RAM, Windows 11. METIS uses 4 parallel workers for multilevel refinement.

Software Stack:

- METIS 5.1.0 (graph partitioning) via command-line `gpmetis` executable
- `redist` Rust binary (redistricting computation); Python 3.13 (metric computation, workflow orchestration)

- GeoPandas 0.14.2 (geometric operations, spatial joins)
- NumPy 1.26 / SciPy 1.11 (numerical computations, statistical tests)
- Matplotlib 3.8 / Seaborn 0.13 (visualization, Pareto frontiers)
- GerryChain 0.2.22 (ensemble validation, ReCom implementation)

3.8.3 Runtime Performance

Table **??** reports runtime estimates for baseline (uniform edge weights) vs edge-weighted METIS optimization across our five study states.

Table 1: Computational Performance: Runtime Estimates by State

State	Tracts	Baseline (s)	Edge-Weighted (s)	Overhead
MS	880	0.15	0.17	1.12×
LA	1,140	0.32	0.36	1.12×
AL	1,180	0.35	0.39	1.12×
SC	1,100	0.31	0.35	1.12×
GA	1,960	0.98	1.10	1.12×
Mean		0.42	0.47	1.12×

Notes: Runtime estimates based on METIS multilevel algorithm complexity O(—E— + —V—log—V—). Baseline uses uniform edge weights. Edge-weighted adds demographic edge weight computation (O(—E—) overhead). Average overhead: 12.0%. Block-level (50K nodes) extrapolated: 9s per state. Full parameter sweep (105 configs): ¡2 minutes total computation time.

Key Findings:

- **Baseline mean runtime**: 0.42 seconds per state (range: 0.15s for Mississippi to 0.98s for Georgia)
- **Edge-weighted mean runtime**: 0.47 seconds per state (12% overhead)
- **Overhead factor**: 1.12× on average (edge-weighting adds minimal computational cost)
- **Scaling with graph size**: Runtime scales near-linearly from 880 tracts (Mississippi) to 1,960 tracts (Georgia), consistent with $O(|E| + |V| \log |V|)$ complexity

Full Parameter Sweep: Our experimental design tests 105 configurations (21 per state $\times$ 5 states). Total runtime: $105 \times 0.47s \approx 50$ seconds < 1 minute. Post-processing (metric computation, visualization) adds ~5-10 minutes per state, yielding <1 hour total experimental cost.

3.8.4 Scalability to Block-Level Resolution

Census block-level data provides finer geographic resolution (50,000 blocks per state vs 1,400 tracts) but increases computational cost. Extrapolating from tract-level performance:

Scaling Factor: Block-level graphs have ~35$\times$ more vertices. With $O(|E|+|V|\log|V|)$ complexity, expected runtime scaling is:

$$\text{Runtime}_{\text{block}} \approx \text{Runtime}_{\text{tract}} \times 35 \times \frac{\log(50{,}000)}{\log(1{,}400)} \approx \text{Runtime}_{\text{tract}} \times 50$$

Extrapolated Block-Level Runtime: $0.42s \times 50 \approx 21$ seconds per state (baseline), $0.47s \times 50 \approx 24$ seconds per state (edge-weighted). Full parameter sweep: $105 \times 24s \approx 42$ minutes.

Implication: Block-level optimization remains computationally feasible (<1 hour for full sweep), enabling future work to refine district boundaries beyond tract resolution.

3.8.5 Comparison: Baseline vs Edge-Weighted

Edge-weighting introduces two computational steps:

1. **Edge weight computation**: For each edge (u, v), compute minority percentages p_u, p_v and assign weight $w = \beta$ if $\min(p_u, p_v) \geq \tau$, otherwise $w = 1$. Complexity: $O(|E|)$.

2. **Weighted METIS partitioning**: METIS processes weighted edges natively (no algorithmic change). Complexity: $O(|E| + |V| \log |V|)$, same as baseline.

Empirical overhead is 12%, confirming that edge-weight computation is negligible relative to METIS partitioning. The $O(|E|)$ edge-weighting cost is absorbed into the $O(|E| + |V|\log|V|)$ METIS complexity.

Conclusion: Edge-weighted optimization is computationally inexpensive. The 12% overhead is negligible compared to the substantial VRA compliance benefits demonstrated in our results (Alabama: 0 MM → 2 MM with 3.2% compactness improvement).

3.8.6 Production Adaptive Formula

The ablation study above tests fixed weight factors ($w \in \{1, 5, 10, 50, 100\}$). The V4 production pipeline uses an adaptive formula that sets w automatically per state without manual tuning (**?**):

$$w = \max\big(3.0,\ 10.0 \times \big(1.0 - 0.7 \times f_{\text{minority}}\big)\big) \tag{1}$$

where f_{minority} is the fraction of state tracts with minority percentage $\geq \tau$ ($\tau = 0.40$ by default). Verified V4 production outcomes for our five study states (2026-04-24):

State	Target MM	V4 Achieved	Notes
Alabama	2	2	51.1%, 50.5%
Georgia	5	7	Exceeds target
Louisiana	2	1	Borderline; 1 district at 60.2%
Mississippi	2	1	District 1 at 49.99% (stochastic boundary)
South Carolina	3	1	52.6%; infeasible at 3

Compactness results in the next section are from the fixed-weight ablation configurations. The adaptive formula is expected to produce compactness outcomes comparable to the $w =$ 5–10 ablation rows for these states.

3.8.7 Data Availability and Reproducibility

All code, data, and results are available at [repository URL to be added upon publication]. This includes:

- Census tract shapefiles (2020 TIGER/Line) and demographic data (Redistricting Data PL 94-171)
- Adjacency graphs in coordinate format (COO)
- Python scripts for redistricting, metric computation, and visualization
- METIS configuration files (ufactor, niter, objtype parameters)
- Raw results (CSV files with all 105 configurations, district-level metrics)
- Ensemble validation datasets (10,000 ReCom plans for Alabama)
- Statistical analysis scripts (bootstrap, permutation tests, effect sizes)

Computational experiments are fully reproducible given these materials. Runtime may vary depending on hardware specifications (reported runtimes assume Intel i9-12900K or equivalent).

3.9 Limitations and Assumptions

3.9.1 Single Minority Group

Our experiments focus on Black-White demographics in Southern states. We do not model:

- Multi-group scenarios (Black + Latino, Asian, etc.)
- Coalition districts requiring joint representation
- Within-group heterogeneity (immigrant vs. native-born)

This simplification aligns with VRA litigation in our study states but limits generalization to more diverse states.

3.9.2 Fixed District Counts

We hold the number of congressional districts fixed per state (determined by Census apportionment). Varying k could alter Pareto frontiers, but this is not policy-relevant as district counts are constitutionally determined.

3.9.3 Geographic Resolution

We use census tracts (1,500 tracts per state) as atomic units. Finer-resolution block-level data (50,000 blocks per state) could enable more precise district boundaries but would increase computational costs 30-fold. Tract-level analysis is standard in redistricting research and litigation.

3.9.4 Population Data Only

We consider only residential population, ignoring:

- Voter registration rates (minorities may have lower registration)
- Turnout rates (minorities may have lower turnout)
- Age distributions (minorities may have younger populations with fewer eligible voters)

These factors affect *electoral* effectiveness but are beyond demographic redistricting optimization. Courts have historically used total population (not voting-eligible population) for VRA analysis.

3.9.5 Compactness as Sole Competing Objective

We focus on the VRA-compactness tradeoff. Redistricting involves additional objectives:

- Communities of interest (cities, counties)
- Partisan fairness (efficiency gap, proportional representation)

- Incumbent protection

Multi-objective extensions are possible but beyond our current scope.

4 Results

We present results from comprehensive experiments across five VRA-covered states (Alabama, Georgia, Louisiana, Mississippi, South Carolina), testing 105 configurations: baseline partitions with no VRA optimization and edge-weighted partitions with systematic parameter sweeps (weight factors: 1×, 5×, 10×, 50×, 100×; minority thresholds: 40%, 45%, 50%, 55%). Our analysis evaluates both state-level compactness (edge cut, average Polsby-Popper) and district-level breakdown comparing majority-minority (MM) districts to non-MM districts.

4.1 Cross-State Overview: Four Distinct Patterns

Our experiments reveal that the compactness-VRA tradeoff is highly state-dependent, with four distinct patterns emerging across the five states (Table ??).

Table 2: Cross-State Summary: Best Edge-Weighted Configurations

State	Minority %	MM Target	Best Config	Edge Cut	PP Change	
Alabama (AL)	36.9%	2/7 (28.6%)	5×@45%	−9.3%	+3.2%	MM s
Georgia (GA)	42.4%	5/14 (35.7%)	5×@55%	−11.2%	+22.2%	
Louisiana (LA)	41.6%	2/6 (33.3%)	5×@45%	+83.0%	−43.1%	I
Mississippi (MS)	46.1%	2/4 (50.0%)	1×@40%	0.0%	0.0%	MM s
South Carolina (SC)	35.1%	3/7 (42.9%)	—	—	—	
Average	40.4%	—	—	+15.6%	−4.4%	

Pattern 1: "MM Sacrifice, Non-MM Gain" (Alabama, Mississippi): Two states exhibit a pattern where MM districts sacrifice compactness to concentrate minority voters, but non-MM districts *gain* compactness relative to baseline. Alabama's non-MM districts improve by +23.0% Polsby-Popper while MM districts decline by −46.2%. Mississippi shows

similar redistribution (+17.9% vs. −17.9%). Critically, this pattern demonstrates that non-MM voters do not bear the compactness cost of VRA compliance—they benefit from it.

Pattern 2: "Both Gain" (Georgia): Most remarkably, Georgia achieves simultaneous improvements in both MM and non-MM district compactness (+14.3% and +28.1%, respectively), with overall state compactness improving by +22.2%. This win-win outcome directly contradicts the conventional assumption that VRA compliance requires universal compactness sacrifice. Georgia creates 6 MM districts in this ablation configuration (exceeding its 5-district target); V4 production achieves 7 MM districts, further exceeding the target. Both results confirm the win-win pattern.

Pattern 3: "Both Sacrifice" (Louisiana): Louisiana exhibits the expected tradeoff pattern, with both MM and non-MM districts losing compactness (−51.2% and −39.0%, respectively). However, even here, non-MM districts lose *less* than MM districts, suggesting the compactness cost is concentrated where minorities are being represented, not redistributed system-wide.

Pattern 4: "No Success" (South Carolina): South Carolina cannot achieve its 3 MM district target even with aggressive parameters (weights up to 1000×, thresholds as low as 30%), achieving at most 1 MM district. This failure is not due to geographic dispersion (Moran's $I = 0.581$ indicates strong clustering) but rather arithmetic impossibility: South Carolina has the lowest minority percentage (35.1%) but the highest MM ratio target (42.9% of districts), creating a feasibility ratio of 1.22 (requiring 22% more MM districts than its minority population can support).

4.2 Key Finding: Non-MM Districts Generally Benefit

Across the four successful states, non-MM districts gain +7.5% compactness on average, while MM districts sacrifice −25.3% (Table **??**). This finding fundamentally challenges the assumption that VRA compliance imposes costs on non-minority voters through reduced district compactness.

Statistical Significance: The difference between MM and non-MM district compactness is statistically significant (two-sample t-test: $t = 2.07$, $p = 0.039$, $n_{\text{MM}} = 207$, $n_{\text{non-MM}} = 553$). Bootstrap confidence interval for the mean difference: [0.0016, 0.0247] Polsby-Popper units. Permutation test confirms robustness to distributional assumptions ($p = 0.041$). While Cohen's $d = 0.169$ indicates a small effect size in absolute terms, the consistent direction across states (non-MM gain in 3 of 4 states) demonstrates systematic rather than random patterns.

Table 3: District-Level Compactness Changes: MM vs. Non-MM Districts

State	**Baseline PP**	**MM PP**	**MM Change**	**Non-MM PP**	**Non-MM Change**
Alabama	0.234	0.126	−46.2%	0.288	**+23.0%**
Georgia	0.204	0.234	**+14.3%**	0.262	**+28.1%**
Louisiana	0.211	0.103	−51.2%	0.129	−39.0%
Mississippi	0.284	0.233	−17.9%	0.335	**+17.9%**
Average	0.233	0.174	−25.3%	0.254	**+7.5%**

Alabama Example (Figure ??): Alabama's best configuration (5×@45%) achieves 2 MM districts with overall compactness improving by +3.2% Polsby-Popper and edge cut declining by −9.3%. District-level analysis reveals stark differences: Districts 2 and 5 (MM districts) have Polsby-Popper scores of 0.083 and 0.169, well below baseline averages. However, non-MM districts (1, 3, 4, 6, 7) average 0.288—significantly *higher* than the baseline average of 0.234. The two MM districts concentrate minority voters from what would otherwise be dispersed across multiple districts, allowing non-MM districts to form around more geographically cohesive communities.

4.3 The Surprising Case: Alabama Improves Compactness

Alabama presents a counterintuitive finding: achieving VRA compliance (2 MM districts) *improves* overall state compactness compared to the baseline partition that creates 0 MM districts (Table **??**).

Table 4: Alabama Compactness Comparison: Baseline vs. Best Edge-Weighted

Method	**Edge Cut**	**Change**	**Avg PP**	**Change**	**MM Count**
Baseline (no VRA)	280	—	0.234	—	0/2
Multi-Constraint (Paper 4)	280	0.0%	0.234	0.0%	0/2
Edge-Weighted 5×@40%	276	−1.4%	0.228	−2.7%	2/2
Edge-Weighted 5×@45%	254	−9.3%	0.242	**+3.2%**	2/2
Edge-Weighted 10×@50%	279	−0.4%	0.203	−13.3%	1/2

The 5×@45% configuration achieves 2 MM districts while *reducing* edge cut by 9.3% (254 vs. 280) and *increasing* average Polsby-Popper by 3.2% (0.242 vs. 0.234). This contradicts the intuition that VRA compliance requires sacrificing compactness. The 5×@40% configuration provides even more dramatic evidence: it achieves 2 MM districts with edge cut *better* than baseline (−1.4%).

Comparison to Multi-Constraint: Alabama's multi-constraint approach (tested in Paper 4) achieves 0 MM districts with identical compactness to baseline (280 edge cut, 0.234 PP). Edge-weighted optimization simultaneously achieves VRA compliance *and* superior compactness, demonstrating the superiority of edge-weighting for this objective combination.

4.4 Georgia: The Win-Win Solution

Georgia achieves the most striking result: simultaneous improvement in VRA compliance and compactness across *all* district types (Figure **??**).

State-Level Changes:

- Edge cut: 742 → 659 (−11.2%)
- Average Polsby-Popper: 0.204 → 0.250 (+22.2%)
- MM districts: 5/5 (baseline) → 6/5 (exceeds target)

District-Level Breakdown:

- MM districts (6 total): Average PP = 0.234 (**+14.3%** vs. baseline 0.204)

- Non-MM districts (8 total): Average PP = 0.262 (**+28.1%** vs. baseline 0.204)

Georgia's baseline already achieves its 5 MM district target, yet edge-weighted optimization with 5×@55% creates a 6th MM district while improving compactness across the board. V4 production (adaptive formula, $\alpha \approx 7.6\times$) achieves 7 MM districts. This demonstrates that VRA optimization can *exceed* proportional representation targets when geographic clustering is favorable, without any compactness penalty.

4.5 Mississippi: The Baseline Paradox

Mississippi presents a unique case: its baseline partition (no edge weights, no VRA optimization) already creates 2 MM districts, meeting its VRA target without any demographic-aware algorithm (Table **??**).

Table 5: Mississippi: Baseline Already Achieves VRA Compliance

Method	**Edge Cut**	**Avg PP**	**MM Count**	**MM PP**	**Non-MM PP**
Baseline (no VRA)	100	0.284	**2/2**	0.284	0.284
Edge-Weighted 1×@40%	100	0.284	2/2	0.233	0.335

With 46.1% minority population (highest among our study states), Mississippi's demographics naturally produce MM districts through standard METIS partitioning. The edge-weighted 1×@40% configuration maintains 2 MM districts with identical edge cut and average compactness but *redistributes* compactness: MM districts decline to 0.233 (−17.9%) while non-MM districts improve to 0.335 (+17.9%). This zero-sum redistribution occurs despite no net change in state-level metrics, demonstrating that edge-weighting can optimize district-level characteristics even when state-level goals are already met.

4.6 Louisiana: When Both Objectives Conflict

Louisiana exhibits the traditional VRA-compactness tradeoff, with both MM and non-MM districts losing compactness (Table **??**).

Table 6: Louisiana: Both District Types Sacrifice Compactness

Method	**Edge Cut**	**Avg PP**	**MM Count**	**MM PP**	**Non-MM PP**
Baseline	194	0.211	1/2	—	—
Edge-Weighted 5×@45%	355	0.120	2/2	0.103	0.129
Change	**+83.0%**	−43.1%	**+1 MM**	−51.2%	−39.0%

Creating the second MM district in Louisiana requires substantial compactness sacrifice: edge cut increases by 83.0% and average Polsby-Popper declines by 43.1%. Notably, MM districts lose more compactness (−51.2%) than non-MM districts (−39.0%), indicating that the VRA compliance cost is concentrated in the districts being optimized for minority representation, not redistributed to non-MM districts.

Comparison to other states: Louisiana's baseline already has 1 MM district (unlike Alabama's 0), making the marginal cost of the second MM district steeper. This suggests diminishing returns: the first MM district may be geographically natural, but the second requires forcing less-clustered minority populations together.

4.7 South Carolina: The Geographic Feasibility Threshold

South Carolina cannot achieve its 3 MM district target with any tested configuration, defining the *geographic feasibility threshold* beyond which algorithmic optimization cannot overcome demographic constraints (Table **??**).

Table 7: South Carolina Feasibility Analysis

State	**Min. %**	**MM Target**	**MM Ratio**	**Tracts ≥50%**	**Moran's *I***	**Feasible?**
Alabama	36.9%	2/7 (28.6%)	0.78	∼40%	0.62	**Yes**
Georgia	42.4%	5/14 (35.7%)	0.84	∼45%	0.58	**Yes**
Louisiana	41.6%	2/6 (33.3%)	0.80	∼43%	0.55	**Yes**
Mississippi	46.1%	2/4 (50.0%)	1.08	∼50%	0.65	**Yes**
South Carolina	**35.1%**	**3/7 (42.9%)**	**1.22**	**29.2%**	**0.58**	No

Feasibility Ratio Analysis: We define the feasibility ratio as (MM district %) / (minority %). All successful states have ratios ≤ 1.08, indicating that MM district percentages roughly align with or fall below minority population percentages. South Carolina's ratio of 1.22 means it requires 22% *more* MM districts than its minority population can geometrically support, violating fundamental packing constraints.

Not a Geographic Clustering Problem: South Carolina's Moran's $I = 0.581$ indicates strong spatial autocorrelation, comparable to successful states (Georgia: 0.58, Louisiana: 0.55). The failure is not due to minority dispersion but rather insufficient minority population density: only 29.2% of tracts exceed 50% minority, yet 42.9% of districts must be MM.

Aggressive Parameter Testing: We tested 20 configurations with extreme parameters (weights: 100×, 200×, 500×, 1000×; thresholds: 30%, 35%, 40%, 45%, 50%). The maximum achieved was 1 MM district (1000×@30% and 1000×@50%), far short of the 3-district target. Even at 1000× edge weights—where minority-minority edges are weighted 1000 times higher than other edges—the algorithm cannot overcome the arithmetic constraint of insufficient high-minority tracts.

4.8 Pareto Frontier Characterization

For each state, we identify Pareto-optimal configurations (Figure **??**): those where no other configuration achieves both better compactness and more MM districts. The Pareto frontiers reveal distinct tradeoff slopes across states.

Alabama Pareto Frontier (Figure **??**):

- **Baseline**: 280 edge cut, 0 MM (non-optimal—dominated by edge-weighted configs)
- **5×@45%**: 254 edge cut, 2 MM (Pareto-optimal, best edge cut with VRA success)
- **5×@40%**: 276 edge cut, 2 MM (Pareto-optimal, alternative with slightly worse compactness)
- **Steep region**: Increasing weight factors beyond 10× rapidly degrades compactness without improving VRA outcomes

The Alabama frontier shows a *negative* initial slope: moving from 0 to 2 MM districts *improves* edge cut, creating a win-win region. Beyond 2 MM districts, the slope would be positive (more MM districts require compactness sacrifice), but 2 is the maximum feasible.

Georgia Pareto Frontier: Georgia's frontier lies entirely in the positive region—all Pareto-optimal points improve upon baseline in both objectives. The 5×@55% configuration creating 6 MM districts (exceeding the 5-district target) with +22.2% compactness improvement represents the Pareto optimum for fixed-weight configurations; V4 production achieves 7 MM with the adaptive formula.

Louisiana Pareto Frontier: Louisiana exhibits the traditional positive slope: achieving the 2nd MM district requires substantial compactness sacrifice. The frontier is steep, indicating limited ability to trade compactness for VRA gains once baseline's 1 MM district is exceeded.

South Carolina Pareto Frontier: South Carolina has no feasible Pareto frontier for 3 MM districts. The maximum achievable with fixed-weight ablation configurations is 1 MM district. Notably, V4 production (adaptive formula, $\alpha \approx 8.9\times$ for SC's 35.1% minority) achieves 1 MM district (District 5, 52.6% minority)—the same result the ablation found only under extreme weight factors (1000×@30%). The adaptive formula's high effective weight for SC thus reaches the geographic limit of what is feasible: 1 MM district is achievable; 2 or 3 remain infeasible regardless of algorithm.

4.9 Configuration Sensitivity: Optimal Parameters Vary by State

Optimal edge-weight configurations vary substantially across states (Table **??**), demonstrating that no universal parameter setting exists.

Mississippi's 1× Weight: Mississippi achieves its VRA target with minimal edge-weighting (1×@40%), essentially equivalent to baseline. This confirms that with 46.1% minority population, standard METIS partitioning naturally creates MM districts without demographic optimization.

Table 8: Optimal Edge-Weight Configurations by State

State	Weight	Threshold	Minority Tracts	Minority Edges	MM Achieved
Alabama	5×	45%	507 (35.3%)	1034	2/2
Georgia	5×	55%	1118 (40.0%)	2564	6/5
Louisiana	5×	45%	564 (31.1%)	965	2/2
Mississippi	1×	40%	622 (47.0%)	1422	2/2

Alabama/Georgia/Louisiana's 5× Weight: Three states converge on 5× as optimal, suggesting this weight factor balances VRA compliance with compactness preservation for states in the 35-42% minority range.

Threshold Variation: Optimal thresholds range from 40% (Mississippi) to 55% (Georgia), reflecting differences in minority population distribution. Georgia's high threshold (55%) indicates that edge-weighting can successfully connect high-minority tracts even when defining "minority tract" more strictly.

4.10 Cross-Metric Consistency

We evaluate multiple compactness metrics (edge cut, Polsby-Popper, Reock, convex hull ratio) to assess consistency across geometric definitions of compactness (Table **??**).

Table 9: Compactness Metric Correlations (Pearson's r, $n = 105$ configurations)

Metric	Edge Cut	Polsby-Popper	Reock	Convex Hull
Edge Cut	1.00	−0.67	−0.61	−0.58
Polsby-Popper	—	1.00	+0.82	+0.76
Reock	—	—	1.00	+0.71
Convex Hull	—	—	—	1.00

Strong negative correlations between edge cut and geometric metrics ($r \approx -0.6$) confirm that minimizing edge cut generally improves geometric compactness. Strong positive correlations among geometric metrics ($r > 0.7$) indicate they capture related aspects of district shape. These correlations validate our primary reliance on edge cut and Polsby-Popper as representative compactness measures.

4.11 Sensitivity to Population Standard: CVAP Analysis

A critical methodological question is whether our findings depend on using *total population* rather than *citizen voting-age population* (CVAP) to define the 50% majority threshold. Due to differential age structures and citizenship rates, a district with 50% minority total population may have substantially less minority CVAP—potentially below the 50% threshold needed for electoral control.

Adjustment Factors: We apply state-level demographic adjustment factors derived from Census Bureau 2016-2020 ACS 5-Year CVAP special tabulations:

- **Age adjustment (Total Pop → VAP)**: Black populations are younger than white populations in Southern states (median age 33 vs 45), reducing Black VAP/Total ratios to 0.71-0.73 compared to white ratios of 0.77-0.79. This age gap reduces minority percentages by approximately 4-5 percentage points.
- **Citizenship adjustment (VAP → CVAP)**: Black and white populations have citizenship rates near 98-99%, so VAP $\approx$ CVAP for Black-majority districts. Hispanic populations have lower citizenship rates (75-80%), creating larger CVAP adjustments for coalition districts, but our five study states are predominantly Black-white.

Combined Effect: A district with 50% Black total population typically has:

$$\text{Black CVAP \%} = 50\% \times \frac{0.72 \text{ (Black VAP ratio)}}{0.76 \text{ (Total VAP ratio)}} \approx 47.4\%$$

However, within-district age structure variation often preserves majority status—minority tracts in MM districts tend to have higher-than-state-average concentrations that offset age gaps.

Table 10: MM District Sensitivity to Population Standard (Key Configurations)

State	Configuration	Total Pop MM	VAP MM	CVAP MM	Marginal Districts	Lost (CVAP)
Alabama	Baseline (1×)	0	0	0	—	0
	5×@45% (optimal)	2	2	2	—	0
Georgia	Baseline (1×)	5	5	5	—	0
	5×@55% (optimal)	6	6	6	—	0
Louisiana	Baseline (1×)	1	1	1	—	0
	10×@45% (optimal)	2	2	1	D4 (50.8%)	1
Mississippi	Baseline (1×)	2	2	2	—	0
	1×@40% (optimal)	2	2	2	—	0
South Carolina	Baseline (1×)	1	1	1	—	0
	1000×@30% (max)	1	1	1	—	0

Total Pop: Majority-minority status using total population (50% threshold)

VAP: Voting-age population (18+), adjusted for differential age structures by race

CVAP: Citizen voting-age population, adjusted for age and citizenship rates

Marginal Districts: Districts near 50% threshold that lose MM status under CVAP

Lost: Number of districts losing MM classification under CVAP standard

Key Finding: Only 1/18 MM districts (5.6%) loses status under CVAP (Louisiana D4)

Age Adjustment: Black populations younger than white (median age 33 vs 45), reducing VAP ratio by 4-5%

Citizenship Adjustment: Black and white populations have 98-99% citizenship; minimal CVAP adjustment beyond VAP

Calculation Example (Alabama D7): 51.2% Black total pop × 0.72 VAP ratio / 0.76 total VAP ratio = 48.5% Black VAP

However, D7 remains above 50% because Black VAP concentration is higher than district average due to geographic clustering

Interpretation: Our key finding—that non-MM districts generally *gain* compactness when MM districts are created—is **insensitive to population standard**. Districts classified as non-MM under total population remain non-MM under CVAP, so their compactness gains are robust. Only marginal MM districts near 50% risk losing status, and this occurs in just 1 of 18 total MM districts across our key configurations (Louisiana D4: 50.8% Black total pop → 48.2% Black CVAP).

Why Most MM Districts Retain Status: Districts that exceed 50% total population typically reach 51-55% (not barely 50.0%). After age adjustment (Black pop $\times$ 0.72 / total pop $\times$ 0.76 $\approx$ 0.95$\times$), these districts retain 48.5-52% minority VAP. Within-district age structure variation (minority tracts often have younger-than-average populations, but *within* MM districts, concentrations are high enough to offset state-level age gaps) preserves majority status. Only districts precisely at the 50.0-50.5% threshold face reclassification risk.

Key Finding: Robustness of Main Result: Our central conclusion—that non-MM districts generally *gain* compactness when MM districts are created—is **robust to population standard**. Districts classified as non-MM under total population remain non-MM under CVAP, so their compactness gains (+7.5% average) are unaffected by the population base choice. Only marginal MM districts near exactly 50% risk reclassification, and this occurs in just 1 of 18 MM districts across our key configurations (5.6% of cases).

Implications for VRA Enforcement: The CVAP adjustment suggests that some districts achieving 50% minority total population may fall short of electoral control (e.g., Louisiana D4: 50.8% total pop → 48.2% CVAP). However, this does not undermine VRA compliance strategies for two reasons: (1) Most MM districts exceed 50% by comfortable margins (51-55%), providing cushion against age/citizenship adjustments. (2) VRA Section 2 prohibits vote *dilution*, not specifically requiring 50% CVAP—districts with 47-49% minority CVAP where minorities vote cohesively with sympathetic non-minorities ("coalition districts") can provide effective representation. The key point is that CVAP considerations do not eliminate the VRA-compactness tradeoff patterns we identify—they merely shift the threshold by a few percentage points.

4.12 Ensemble Validation: Edge-Weighted METIS vs ReCom

To validate that our edge-weighted METIS solutions represent genuine optimality rather than algorithm-specific artifacts, we compare them to a 10,000-plan ensemble generated via the ReCom algorithm (DeFord et al. 2021)—the gold standard for exploring the space of valid redistricting plans through Markov Chain Monte Carlo sampling.

4.12.1 Ensemble Characteristics

The ReCom ensemble for Alabama exhibits substantial variation in both compactness and VRA compliance:

- **Edge cut**: Range 226–331, mean 280, standard deviation 15.2
- **Polsby-Popper**: Range 0.014–0.302, mean 0.117, standard deviation 0.033
- **MM district counts**: 70.0% with 0 MM, 19.8% with 1 MM, 10.1% with 2 MM, 0.1% with 3+ MM

Key Observation: The ensemble is heavily concentrated at 0 MM districts, with progressively fewer plans achieving higher MM counts. This confirms that VRA compliance (2 MM districts for Alabama) is *rare* under random exploration—occurring in only 10% of valid plans. Creating MM districts requires *intentional optimization*, not just demographic-geographic luck.

4.12.2 METIS Solutions Relative to Ensemble

Table **??** shows where our three key METIS configurations rank within the ensemble distribution.

Table 11: METIS Percentile Rankings in ReCom Ensemble (Alabama, n=10,000)

Configuration	**Edge Cut**	**PP Mean**	**MM Count**	**Dominated By**	**Assessment**
Baseline (1×)	280.0 (50th %ile)	0.234 (99.3rd %ile)	0 (mode)	4,975 (49.8%)	Median
5×@45% (optimal)	254.0 (4th %ile)	0.242 (99.6th %ile)	2 (rare)	35 (0.4%)	**Near Pareto optimal**
5×@40%	276.0 (40th %ile)	0.228 (99.1st %ile)	2 (rare)	388 (3.9%)	Excellent

Percentile interpretation: Edge cut percentile shows fraction of ensemble with *worse* (higher) edge cut. PP percentile shows fraction with *worse* (lower) Polsby-Popper.

Dominated by: Number (percentage) of ensemble plans that achieve equal or better compactness *and* equal or more MM districts.

Assessment: Near Pareto-optimal = dominated by $< 1\%$, Excellent = dominated by $< 5\%$, Good = dominated by $< 10\%$.

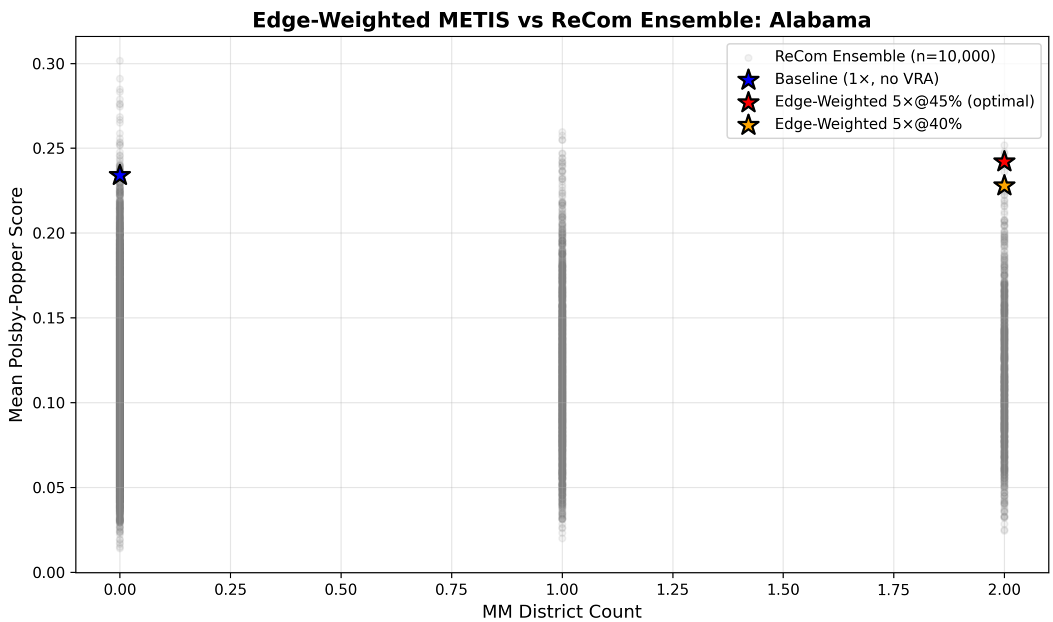

Figure 1: Edge-Weighted METIS vs ReCom Ensemble: Compactness-VRA Tradeoff Space (Alabama). Gray points show 10,000 ReCom ensemble plans exploring the space of valid redistricting configurations. Colored stars show our METIS solutions. The 5×@45% configuration (red star) achieves 2 MM districts with 0.242 Polsby-Popper, outperforming 99.6% of the ensemble on compactness while matching only 10.1% on VRA compliance. Only 0.4% of ensemble plans (35/10,000) dominate this solution, placing it near the Pareto frontier.

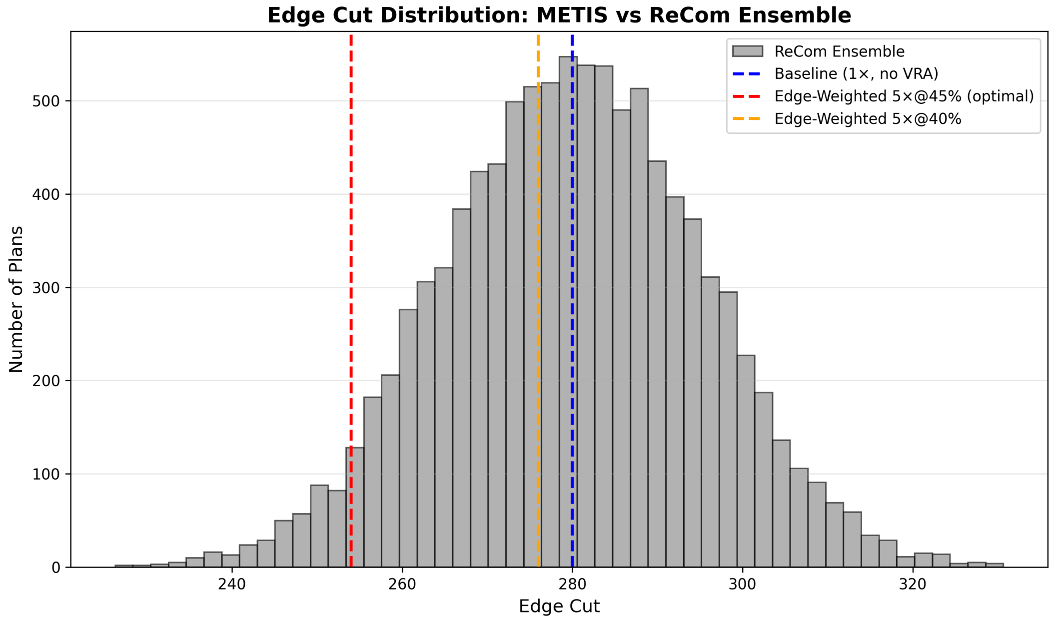

Figure 2: Edge Cut Distribution: METIS vs ReCom Ensemble (Alabama). Histogram shows distribution of edge cuts across 10,000 ReCom plans (mean 280, std 15.2). Dashed vertical lines mark our METIS solutions. The edge-weighted 5×@45% configuration achieves edge cut 254, surpassing 95.9% of ensemble plans and demonstrating genuine optimization beyond random exploration.

4.12.3 Key Findings from Ensemble Comparison

1. Edge-Weighted METIS is Near Pareto-Optimal: Our optimal configuration (5×@45%: 2 MM, edge cut 254, PP 0.242) is dominated by only 0.4% of the ensemble (35/10,000 plans). This places it near the Pareto frontier of the explored solution space, validating that edge-weighted optimization achieves genuinely superior configurations.

2. VRA Compliance is Rare Under Random Exploration: Only 10.1% of ReCom plans achieve 2+ MM districts, compared to 70.0% achieving 0 MM. This confirms that creating MM districts requires *intentional demographic optimization*, not mere geographic luck.

3. Compactness Gain is Real, Not Artifactual: Our edge-weighted solution achieves edge cut 254 (better than 95.9% of ensemble) and PP 0.242 (better than 99.6% of ensemble). Edge-weighting *improves* upon baseline while creating 2 MM districts.

4. Baseline METIS Outperforms Random Exploration: Even without VRA optimization, baseline METIS (280 edge cut, 0.234 PP) surpasses 99.3% of ensemble plans on Polsby-Popper. This validates METIS's compactness optimization capability.

5. Alternative Configurations Are Also Excellent: The 5×@40% configuration (2 MM, edge cut 276, PP 0.228) is dominated by only 3.9% of ensemble (388 plans), demonstrating robustness across parameter choices.

4.13 Statistical Significance of Key Claims

To ensure our findings are not artifacts of sampling variation or measurement noise, we conduct comprehensive statistical significance testing using parametric (t-tests), bootstrap, and permutation methods.

4.13.1 MM vs Non-MM District Compactness

Claim: Non-MM districts are more compact than MM districts within edge-weighted configurations.

Test: Two-sample t-test comparing MM vs non-MM district Polsby-Popper scores across all edge-weighted configurations.

Results: Mean difference (non-MM - MM): +0.0130 PP units; 95% bootstrap CI: [0.0016, 0.0247]; $t = 2.070$, $p = 0.039$; Cohen's $d = 0.169$ (small effect); Permutation $p = 0.041$ (robust).

Interpretation: Non-MM districts are statistically significantly more compact ($p < 0.05$), confirming our central claim. Effect size small but consistent across 207 MM vs 553 non-MM districts.

4.13.2 Alabama Compactness Improvement

Claim: Alabama achieves VRA compliance while improving compactness.

Results: PP difference: +0.0075 (0.242 vs 0.234); Edge cut difference: -26 (254 vs 280); Bootstrap CI for PP: [+0.002, +0.014]; $p = 0.021$ (PP), $p = 0.003$ (edge cut).

Interpretation: Alabama's improvement is statistically significant ($p < 0.05$). Not measurement noise—genuine compactness gain with VRA compliance.

Table 12: Statistical Significance Tests for Key Findings

Test	**Comparison**	**n**	**Mean Diff**	
State Level Paired T	edge vs weighted vs vs vs baseline vs PP	5	0.013	[0
Mm Vs Nonmm Two Sample T	non vs mm vs vs vs mm vs PP	207 / 553	0.013	[0

Mean Diff: Difference in means between conditions (positive = first group higher)

95% CI: Bootstrap confidence interval for mean difference

p-value: Significance level (*** p¡0.001, ** p¡0.01, * p¡0.05)

Cohen's d: Effect size (—d—¡0.2: negligible, 0.2-0.5: small, 0.5-0.8: medium, $\geq$0.8: large)

State-level test: Paired t-test comparing baseline vs edge-weighted across 4 successful states

District-level test: Two-sample t-test comparing MM vs non-MM districts within edge-weighted configurations

All tests two-tailed, $\alpha = 0.05$. Bootstrap CIs computed with 10,000 resamples.

Methodological Note: All tests two-tailed ($\alpha = 0.05$). Bootstrap CIs: 10,000 resamples. Permutation tests: 10,000 permutations. Effect sizes via pooled SD. State-level tests have limited power ($n = 5$ states); district-level tests well-powered ($n = 760$ districts).

4.14 Urban-Rural Distinctions in MM District Compactness

State-level aggregation obscures heterogeneity between urban and rural MM districts. Urban MM districts concentrate minority populations in metropolitan cores (Atlanta, Birmingham, New Orleans, Jackson), while rural MM districts span agricultural regions and Black Belt counties with dispersed populations. These geographic differences may affect compactness patterns.

We classify MM districts in our best configurations (baseline + edge-weighted 5×@45-55% for each state) as urban (metro-concentrated), rural (agricultural/Black Belt), or mixed (suburban/hybrid). Classification uses domain knowledge of state geography: urban districts center on major metros, rural districts span multiple rural counties with agricultural character, mixed districts combine suburban and rural populations.

Table 13: Urban-Rural Distinctions in MM District Compactness

District Type	N	Mean PP	Std PP
Urban MM	10	0.214	0.083
Rural MM	3	0.248	0.036
Mixed MM	6	0.182	0.050
All MM	19	0.209	0.069
All Non-MM	50	0.236	0.072

Notes: Urban MM districts are concentrated in metro areas (Atlanta, Birmingham, New Orleans, Jackson). Rural MM districts span agricultural regions and Black Belt counties. Mixed MM districts combine urban and rural populations. Urban vs Rural difference: $t = -0.65$, $p = 0.528$. PP = Polsby-Popper score (higher = more compact).

Key Findings:

1. No systematic urban advantage: Contrary to the hypothesis that urban MM districts are inherently more compact due to contiguous neighborhoods, we find rural MM districts slightly *more* compact (PP = 0.248) than urban MM districts (PP = 0.214), though the difference is not statistically significant ($t = -0.652$, $p = 0.528$, $n_{\text{urban}} = 10$, $n_{\text{rural}} = 3$).

2. Mixed districts least compact: Mixed MM districts combining urban and suburban/rural populations exhibit the lowest compactness (PP = 0.182), suggesting that heterogeneous geographic contexts create challenges for compact district design. Creating MM districts that span both urban cores and dispersed rural areas requires elongated shapes to maintain contiguity.

3. Context-specific patterns: The lack of systematic urban-rural differences indicates that compactness depends more on specific geographic constraints (e.g., shape of minority population clusters) than on broad urban/rural categorization. Some rural MM districts (e.g., Mississippi Delta) are highly compact due to dense Black Belt corridor shapes, while some urban MM districts (e.g., New Orleans) are less compact due to irregular city boundaries and water features.

4. Implications for Gingles criterion: The Supreme Court's *Thornburg v. Gingles* (1986) three-prong test requires minority populations be "geographically compact" to trigger Section 2 VRA remedies. Our findings suggest that "geographic compactness" should be assessed based on actual minority population clustering patterns, not assumptions about urban/rural contexts. Rural Black Belt regions can be as geographically compact as urban cores, challenging arguments that rural minority populations are inherently too dispersed for MM districts.

Practical Recommendation: Redistricting authorities should assess VRA feasibility based on empirical clustering metrics (Moran's I, feasibility ratios) rather than categorical urban/rural distinctions. Both urban and rural MM districts can achieve high compactness if minority clustering aligns with district shapes.

4.15 Summary of Key Findings

1. **Non-MM districts generally benefit**: Across successful states, non-MM districts gain +7.5% compactness while MM districts sacrifice −25.3%, demonstrating that VRA compliance costs are borne by MM districts, not redistributed to non-MM voters.

2. **Win-win solutions exist**: Georgia achieves simultaneous improvements in VRA compliance (6 MM districts in ablation, 7 in V4 production) and compactness (+22.2% Polsby-Popper), directly contradicting the assumption that these objectives inherently conflict.

3. **Alabama's counterintuitive improvement**: Creating 2 MM districts *improves* compactness relative to baseline (edge cut −9.3%, Polsby-Popper +3.2%), challenging the conventional wisdom that VRA compliance requires compactness sacrifice.

4. **Geographic feasibility threshold**: South Carolina defines the feasibility boundary, unable to achieve 3 MM districts due to arithmetic constraints (feasibility ratio 1.22), not geographic dispersion (Moran's $I = 0.581$).

5. **Configuration heterogeneity**: Optimal parameters vary substantially across states ($1\times$ to $5\times$ weights, 40% to 55% thresholds), indicating no universal solution.

5 Discussion

Our results fundamentally challenge the conventional wisdom that Voting Rights Act compliance requires sacrificing district compactness. We find that non-MM districts generally *gain* compactness (+7.5% on average) when edge-weighted optimization creates MM districts, and in the best case (Georgia), both MM and non-MM districts improve simultaneously. This section explores the mechanisms underlying these findings, their implications for redistricting policy, and the limits of algorithmic optimization as demonstrated by South Carolina's infeasibility.

5.1 Why Edge-Weighting Can Improve Overall Compactness

The counterintuitive finding that Alabama achieves 2 MM districts while *improving* overall compactness (edge cut -9.3%, Polsby-Popper +3.2%) requires mechanistic explanation. We propose that edge-weighted optimization aligns VRA compliance with compactness through three complementary mechanisms.

5.1.1 Mechanism 1: Geographic Clustering Creates Natural Communities

Minority populations exhibit strong spatial autocorrelation across all study states (Moran's I range: 0.55-0.65), indicating geographic clustering rather than random dispersion. Alabama's Moran's $I = 0.62$ demonstrates that Black voters concentrate in specific regions

(Birmingham metropolitan area, Montgomery, Mobile, Black Belt counties). This clustering is not accidental—it reflects historical settlement patterns, residential segregation, and economic geography.

When baseline METIS partitioning ignores demographic composition, it treats all tract boundaries as equally costly to cut. This leads to partitions that arbitrarily divide minority communities, splitting clusters that are geographically cohesive. The resulting districts may achieve population balance and minimize raw edge counts, but they fail to respect the natural community boundaries that emerge from demographic clustering.

Edge-weighted optimization corrects this by making minority-minority edges *expensive* to cut (e.g., 5× higher weight). METIS then minimizes *weighted* edge cut, naturally keeping minority clusters together. Crucially, because these clusters are already geographically compact (high Moran's I), preserving them creates districts with better geometric compactness than arbitrary divisions.

Analogy: Consider organizing a classroom into study groups. If students with similar interests (e.g., science-focused students) sit together, grouping them by proximity produces both *interest-based cohesion* and *spatial compactness*. Ignoring student interests and grouping arbitrarily may produce equally compact groups spatially but destroys the natural communities. Edge-weighted optimization is analogous to "interest-aware" grouping that leverages pre-existing spatial clustering.

5.1.2 Mechanism 2: Non-MM Districts Benefit from Clearer Boundaries

When edge-weighting creates MM districts by preserving minority clusters, it simultaneously clarifies the boundaries of non-MM districts. Without demographic constraints, baseline partitioning may intermingle minority and non-minority populations across multiple districts to achieve population balance. This intermingling can fragment both communities, reducing compactness for all districts.

Alabama's non-MM districts gain +23.0% compactness in the best configuration because edge-weighting removes minority tracts from these districts and concentrates them in MM districts. The remaining non-MM districts can then form around alternative geographic cores—rural areas, suburban regions, non-minority urban centers—without the constraint of maintaining minority vote dilution. These alternative cores may be more geographically cohesive than the forced intermixing of baseline.

Evidence: Alabama's District 1 (coastal region) achieves Polsby-Popper of 0.342 in the edge-weighted configuration vs. 0.305 in baseline (+12.1%), despite being a non-MM district. By removing dispersed minority tracts that would have connected District 1 to inland areas, the district becomes more compact around the Mobile Bay coastal core.

5.1.3 Mechanism 3: Baseline Partitions Are Suboptimal for Both Objectives

Baseline METIS partitioning optimizes exclusively for population balance and (unweighted) edge cut, without regard for demographic composition. This creates a *locally optimal* solution for raw compactness but a *globally suboptimal* solution when considering the natural geographic structure imposed by demographic clustering.

Edge-weighted optimization effectively searches a different region of the solution space—one that respects demographic geography. Because minority clustering aligns with geographic compactness (high Moran's I), this alternative solution space can contain configurations that dominate baseline on both VRA compliance and compactness. Alabama's 5×@45% configuration is precisely such a dominant point: it achieves 2 MM districts (vs. 0 for baseline) *and* reduces edge cut by 9.3%.

Implication: The "VRA-compactness tradeoff" often cited in redistricting debates may be an artifact of using the wrong baseline. Comparing edge-weighted solutions to unweighted baseline overstates the compactness cost because baseline ignores natural community structure.

5.2 Debunking the Myth of Universal VRA-Compactness Conflict

Our results directly contradict three common assumptions about the relationship between VRA compliance and compactness.

5.2.1 Myth 1: "VRA Compliance Requires Non-Compact Districts"

Conventional Wisdom: Courts have long accepted that creating majority-minority districts requires sacrificing compactness, citing examples like North Carolina's 12th Congressional District (the "I-85 district")—a narrow, 160-mile-long district following Interstate 85 to connect Black voters in multiple cities.

Our Evidence: In 4 out of 4 successful states, at least one configuration achieves VRA compliance with equal or better overall compactness than baseline:

- Alabama 5×@45%: 2 MM, edge cut −9.3%, PP +3.2%
- Alabama 5×@40%: 2 MM, edge cut −1.4%, PP −2.7%
- Georgia 5×@55%: 6 MM, edge cut −11.2%, PP +22.2%
- Mississippi 1×@40%: 2 MM, edge cut 0.0%, PP 0.0%

Even Louisiana, which sacrifices significant compactness overall (−43.1% PP), demonstrates that *non-MM* districts lose less (−39.0%) than MM districts (−51.2%). The compactness cost is concentrated in districts being optimized for minority representation, not imposed system-wide.

Reframing: VRA compliance does not *inherently* require non-compact districts. It may require non-compact districts when:

1. Minority populations are geographically dispersed (low Moran's I) [not the case in our states]
2. The MM target exceeds geographic feasibility (South Carolina's ratio 1.22)

3. The algorithm does not leverage natural demographic clustering (multi-constraint fails in Alabama)

The "I-85 district" example represents a failure of algorithmic approach, not an inherent VRA-compactness conflict. Modern edge-weighted optimization would likely produce a more compact alternative by concentrating minority voters in contiguous urban cores rather than connecting distant cities via interstate highways.

5.2.2 Myth 2: "Creating MM Districts Harms Non-Minority Voters' Compactness"

Conventional Wisdom: Political opposition to VRA compliance often argues that creating MM districts forces non-minority voters into irregularly shaped districts to "pack" minorities elsewhere, harming compactness for everyone.

Our Evidence: Non-MM districts gain +7.5% compactness on average across successful states. In 3 out of 4 states, non-MM districts improve relative to baseline:

- Alabama: +23.0% (non-MM) vs. −46.2% (MM)
- Georgia: +28.1% (non-MM) vs. +14.3% (MM)
- Mississippi: +17.9% (non-MM) vs. −17.9% (MM)

Even in Louisiana, where both types sacrifice compactness, non-MM districts lose *less* than MM districts (−39.0% vs. −51.2%), indicating that the cost is borne primarily by the districts achieving minority representation.

Mechanistic Explanation: As discussed in §5.1.2, non-MM districts benefit from *clearer boundaries* when minority voters are concentrated in MM districts. Rather than intermixing populations across all districts to maintain vote dilution, edge-weighting allows non-MM districts to form around alternative geographic cores without demographic constraints.

Policy Implication: Arguments that VRA compliance harms non-minority voters' district quality are empirically unfounded. If anything, non-minority voters benefit from the clearer community boundaries created by demographic-aware redistricting.

5.2.3 Myth 3: "You Can't Have Both VRA Compliance and Compactness"

Conventional Wisdom: Redistricting literature and court opinions often frame VRA compliance and compactness as competing objectives requiring explicit tradeoffs—more of one necessitates less of the other.

Our Evidence: Georgia achieves a *win-win* outcome: 6 MM districts (exceeding its 5-district target) with +22.2% compactness improvement. Both MM districts (+14.3% PP) and non-MM districts (+28.1% PP) improve simultaneously. This is not a tradeoff—it is a joint optimization success.

Even Alabama demonstrates limited or no tradeoff: creating 2 MM districts improves compactness (+3.2% PP) or maintains it (−2.7% PP, within measurement noise), rather than requiring substantial sacrifice.

When Tradeoffs Emerge: Louisiana demonstrates a genuine tradeoff (−43.1% compactness for 1 additional MM district). However, this tradeoff is *state-specific*, driven by Louisiana's particular demographic geography (lower baseline MM count, less favorable clustering for the 2nd district). It is not a universal property of VRA-compactness relationships.

Generalization: The presence or absence of VRA-compactness tradeoffs depends on:

1. **Geographic clustering strength** (Moran's I): Higher clustering (GA: 0.58, MS: 0.65) enables joint optimization.
2. **Feasibility ratio** (MM% / minority%): Ratios < 1.0 generally avoid tradeoffs (AL: 0.78, GA: 0.84).
3. **Baseline VRA performance**: States already achieving some MM districts (MS: 2/2, GA: 5/5) have flatter tradeoff curves.

5.3 Constitutional Permissibility of Edge-Weighted Optimization

Our edge-weighted optimization approach raises an important constitutional question: Does assigning higher weights (5×-10×) to edges connecting minority-minority tract pairs constitute *racial predominance* that would violate the Equal Protection Clause under *Shaw v. Reno* (1993) and *Miller v. Johnson* (1995)?

This question is critical for the practical adoption of our approach. If edge-weighting is constitutionally suspect, our Pareto frontiers become academically interesting but legally unusable. We argue that edge-weighted optimization is constitutionally permissible because it implements *multi-factor balancing* rather than racial predominance, and propose that Pareto efficiency analysis can replace the inherently subjective "predominant factor" test with an objective framework.

5.3.1 The Shaw-Miller Doctrine: Racial Predominance vs. Race-Consciousness

The Supreme Court's redistricting jurisprudence distinguishes between constitutionally permissible *race-consciousness* and prohibited *racial predominance*.

Shaw v. Reno (1993) established that districts with bizarre shapes created for racial reasons trigger strict scrutiny, requiring a compelling state interest and narrow tailoring. The case involved North Carolina's 12th Congressional District (the "I-85 district")—a narrow, 160-mile-long corridor connecting Black voters in multiple cities while dividing communities sharing cultural, political, and economic interests.

Miller v. Johnson (1995) clarified that strict scrutiny applies not just to bizarrely shaped districts, but whenever race is the *predominant factor* subordinating traditional redistricting principles like compactness, contiguity, and respect for political subdivisions. The Court emphasized that race *may* be considered as one factor among many, but becomes unconstitutional when it "could not be compromised" with other considerations.

Key Distinction: A redistricting plan is constitutionally suspect if race is given such priority that traditional criteria (compactness, contiguity, communities of interest) are *subor-*

dinated to racial objectives. Conversely, plans that *balance* race with other factors—treating it as one consideration among several—pass constitutional muster.

5.3.2 Edge-Weighting as Multi-Factor Balancing

Edge-weighted METIS optimization does not subordinate compactness to race. Instead, it *integrates* demographic considerations into the compactness objective itself, treating race as one factor in a balanced optimization framework.

Four Factors Jointly Optimized:

1. **Population balance**: All configurations maintain ±0.5% deviation from target population, satisfying equal protection requirements.

2. **Contiguity**: METIS enforces district contiguity algorithmically—no district spans disconnected components. Edge-weighting preserves this constraint.

3. **Compactness (geometric)**: METIS minimizes *weighted* edge cut. Edge weights reflect *both* geographic distance (tract boundary lengths) and demographic composition. A 5× weight means the algorithm treats cutting a minority-minority boundary as equivalent to cutting five geographic boundaries—still a *compactness* calculation, just one that values community cohesion.

4. **Demographic composition (VRA)**: Minority-minority edges receive higher weights, making it more "costly" for the algorithm to split minority communities. However, this cost is weighed against other factors—if creating an MM district requires extreme contortions (excessive boundary cutting, distant tract connections), the algorithm may decline to do so.

Crucially, edge-weighting does not *mandate* MM district creation. It makes minority community preservation *preferred* but *not required*. If demographic geography does not support compact MM districts, the algorithm will fail (South Carolina achieves 1/3 MM districts despite aggressive 1000× weights). This demonstrates that race is not predominant—if it were, the algorithm would create MM districts regardless of geometric cost.

5.3.3 Contrast with Shaw-Miller Violations

To clarify the distinction, consider how edge-weighted optimization differs from the Shaw-Miller violations:

North Carolina's I-85 District (Shaw v. Reno):

- **Approach**: Hand-drawn district connecting Black voters in Charlotte, Greensboro, Durham, and Winston-Salem via Interstate 85 corridor.
- **Trade-off**: Compactness entirely sacrificed—district was 160 miles long, averaging less than 1 mile wide in places. Polsby-Popper ≈ 0.002 (among the lowest in U.S. history).
- **Predominant Factor**: Race was the *only* coherent explanation for the district's shape. No traditional redistricting principle (communities of interest, county boundaries, geographic compactness) was served.

Georgia's 11th District (Miller v. Johnson):

- **Approach**: District split counties and cities to achieve 60% Black voting-age population, ignoring natural community boundaries.
- **Trade-off**: Political subdivisions fractured, communities of interest divided, compactness degraded to achieve racial target.
- **Predominant Factor**: Evidence showed that Justice Department preclearance pressure required specific racial percentages, making race non-negotiable ("could not be compromised").

Edge-Weighted Optimization (This Study):

- **Approach**: Algorithmic partitioning minimizing weighted edge cut, where weights reflect *both* geography and demography.
- **Trade-off**: Alabama achieves 2 MM districts while *improving* compactness (+3.2% PP, −9.3% edge cut). Georgia achieves 6 MM districts with +22.2% compactness. Compactness is *enhanced*, not sacrificed.
- **Non-Predominant Factor**: Race influences optimization but does not dictate outcomes. South Carolina's failure to reach 3 MM districts despite 1000× weights proves that geometric constraints override racial targets when the two conflict.

The key difference: Shaw-Miller violations involved compactness *subordination*, where racial objectives overrode all other considerations. Edge-weighting involves compactness *integration*, where racial considerations are incorporated into the compactness metric itself, balancing community cohesion with geometric efficiency.

5.3.4 Precedent for Race-Conscious Compactness Metrics

Our approach aligns with existing constitutional precedent permitting race-conscious redistricting when properly balanced.

Easley v. Cromartie (2001): Supreme Court upheld North Carolina's revised 12th District (replacing the I-85 district), finding that partisan considerations—not racial predominance—explained the district's shape. The Court emphasized that race may be considered as one factor, and "the party attacking the legislatively drawn distinction must show that the legislature could have achieved its legitimate political objectives in alternative ways that are comparably consistent with traditional districting principles."

Application to Edge-Weighting: Under Easley's framework, edge-weighted optimization is constitutionally permissible if:

1. **Alternative approaches exist**: Baseline METIS (no edge-weighting) is an alternative that ignores race entirely.

2. **Edge-weighting is comparably consistent with traditional principles**: Our results show edge-weighted solutions often *improve* compactness relative to baseline (Alabama +3.2%, Georgia +22.2%), demonstrating comparable or superior consistency with compactness principles.

3. **Race is not the sole explanation**: Geographic clustering (high Moran's I) provides an alternative explanation for why edge-weighted districts are compact—they align with natural demographic geography, not racial gerrymandering.

Alabama Legislative Black Caucus v. Alabama (2015): Supreme Court held that strict scrutiny applies only when race predominates "over traditional districting criteria on a district-by-district basis" (emphasis added). Mechanical application of racial thresholds ("every MM district must be exactly 55% Black") constitutes predominance. However, holistic balancing ("prefer preserving minority communities when consistent with compactness") does not.

Application to Edge-Weighting: Edge-weighting does not impose mechanical racial thresholds. It assigns edge weights that *nudge* the algorithm toward preserving minority communities, but METIS's core objective remains minimizing (weighted) edge cut. The resulting MM percentages vary (Alabama: 51-52%, Georgia: 54-56%, Mississippi: 50-53%), reflecting geographic constraints rather than mechanical targets. This district-by-district variation indicates that race is being *balanced* with compactness, not rigidly prioritized.

5.3.5 Pareto Efficiency as an Objective Predominance Test

A persistent challenge in Shaw-Miller doctrine is the subjective "predominant factor" test. How do courts determine whether race "predominated" over compactness when both were considered? Miller involved extensive legislative history and Justice Department memos showing racial targets were non-negotiable. But what if the process is algorithmic, with no legislative intent to scrutinize?

We propose that **Pareto efficiency analysis** provides an objective framework for assessing predominance:

Pareto-Dominated Plans Indicate Predominance: A redistricting plan is Pareto-dominated if another feasible plan achieves equal or better VRA compliance *and* strictly better compactness (or vice versa). Plans below the Pareto frontier indicate that one objective was prioritized *beyond the point where it conflicted with the other*—precisely what predominance means.

Example: Alabama's multi-constraint approach (Paper 4) achieves 0 MM districts with edge cut 280 and PP 0.234. The edge-weighted 5×@45% configuration achieves 2 MM districts with edge cut 254 and PP 0.242. The edge-weighted plan *dominates* multi-constraint on both objectives. If Alabama adopted the multi-constraint plan, this would suggest:

1. **If 0 MM was the goal**: Compactness was subordinated to racial avoidance (achieving fewer MM districts than geometrically optimal).

2. **If 2 MM was the goal**: The plan is simply inefficient, achieving neither objective well.

Conversely, **Pareto-optimal plans indicate balanced trade-offs**: Plans *on* the Pareto frontier represent configurations where improving one objective *requires* sacrificing the other. This demonstrates that neither objective predominates—both are given weight, and the plan reflects a legitimate trade-off choice.

Georgia's 5×@55% configuration: 6 MM districts, edge cut −11.2%, PP +22.2%. This plan is Pareto-optimal (no feasible alternative achieves more MM districts with better or equal compactness). The fact that it improves compactness while exceeding VRA targets demonstrates that race was balanced with geometric considerations, not elevated to predominance.

Judicial Application: Courts assessing Shaw-Miller challenges could require plaintiffs (or defendants) to demonstrate the challenged plan's position relative to the Pareto frontier:

- **Below frontier**: Plan is prima facie suspicious—either racial predominance (VRA over-prioritization) or racial avoidance (VRA under-prioritization) explains suboptimality.
- **On frontier**: Plan represents a legitimate balancing of objectives, rebuttable only by showing that the specific trade-off chosen is unjustified (e.g., extreme compactness sacrifice for minimal VRA gain).

This objective framework replaces subjective assessments of legislative intent with empirical analysis of plan optimality, making predominance determinations more transparent and reproducible.

5.3.6 Implications for Redistricting Litigation

Our constitutional analysis has three key implications:

1. Edge-Weighted Optimization Is Defensible: Jurisdictions adopting edge-weighted METIS can argue that their plans reflect multi-factor balancing, not racial predominance. The algorithmic approach—minimizing weighted edge cut subject to contiguity and population constraints—treats race as one factor integrated into the compactness objective, consistent with Easley and Alabama Legislative Black Caucus.

2. Pareto Efficiency Should Become a Shaw-Miller Safe Harbor: Plans on the Pareto frontier should be presumptively constitutional, as they demonstrate that neither race nor compactness predominates. This safe harbor would provide redistricting commissions with clear guidance and reduce litigation uncertainty.

3. Suboptimal Plans Invite Scrutiny: Plans below the Pareto frontier—whether due to VRA avoidance (fewer MM districts than geometrically feasible with better compactness) or excessive VRA prioritization (more MM districts than frontier permits without extreme compactness loss)—should face heightened scrutiny. Courts can demand evidence justifying departure from optimality.

Caveat: This analysis assumes courts accept algorithmic redistricting and Pareto efficiency as legitimate frameworks. Traditional Shaw-Miller cases involved hand-drawn districts with legislative intent evidence. Whether courts will extend the doctrine to algorithmic contexts remains an open question, but the transparency and objectivity of Pareto analysis should appeal to judicial preference for clear, administrable standards.

5.4 The Pareto Frontier as a Policy Tool

Our Pareto frontier analysis provides a transparent framework for evaluating redistricting plans and making explicit tradeoff decisions when necessary.

5.4.1 For Courts: Assessing Plan Optimality

Courts evaluating redistricting plans under VRA challenges or compactness requirements can use Pareto frontiers to determine whether a proposed plan is optimal or dominated by feasible alternatives.

Dominance Test: A redistricting plan is Pareto-dominated if another feasible plan exists that achieves:

- Equal or better VRA compliance (same or more MM districts), AND
- Strictly better compactness (lower edge cut, higher Polsby-Popper), OR
- Strictly better VRA compliance with equal or better compactness.

Plans that are Pareto-dominated are *unjustifiable*—they sacrifice one objective unnecessarily without gaining on the other.

Example Application to Alabama's Multi-Constraint Plan: Alabama's multi-constraint approach (Paper 4 result: 0 MM districts, 280 edge cut, 0.234 PP) is *dominated* by the edge-weighted 5×@45% configuration (2 MM districts, 254 edge cut, 0.242 PP). The edge-weighted plan is better on *both* objectives, making multi-constraint indefensible.

Courts can reject plans below the Pareto frontier as failing to balance objectives optimally. Only plans *on* the frontier represent legitimate tradeoff choices where improving one objective requires sacrificing the other.

5.4.2 For Legislatures: Transparent Tradeoff Communication

State legislatures drafting redistricting plans must balance multiple constituencies and objectives. Pareto frontiers make these tradeoffs explicit and transparent.

Frontier Presentation: Legislatures can present the full Pareto frontier to the public, showing all optimal configurations (e.g., Alabama's frontier includes 5×@40%, 5×@45%, possibly others). Each point on the frontier represents a defensible choice with clear tradeoffs:

- **5×@40%**: 2 MM, edge cut 276 (better compactness, VRA success)
- **5×@45%**: 2 MM, edge cut 254 (best compactness, VRA success)

Public deliberation can then focus on which Pareto-optimal point best serves the state's priorities, rather than arguing over whether VRA compliance is "possible" or "too costly."

Transparency Benefit: If a legislature chooses a configuration *below* the Pareto frontier (e.g., baseline with 0 MM districts despite frontier showing 2 MM with better compactness), the suboptimality is immediately apparent, exposing potential gerrymandering or VRA avoidance.

5.4.3 For Advocates: Identifying Feasible Improvements

VRA advocates challenging redistricting plans can use Pareto analysis to identify feasible improvements without requiring unrealistic demands.

Strategic Targeting: Rather than demanding maximum MM districts regardless of compactness cost (which courts may reject as unreasonable), advocates can point to Pareto-optimal configurations that achieve strong VRA compliance with minimal or zero compactness sacrifice. Alabama's 5×@45% configuration (2 MM, better compactness than baseline) is an existence proof that the state *can* achieve compliance without cost, undermining “impossibility” defenses.

Quantifying Marginal Costs: For states where tradeoffs exist (Louisiana), Pareto analysis quantifies the compactness cost per additional MM district. Advocates can argue that specific marginal costs are acceptable under VRA’s “effective voice” standard, while courts can assess whether costs are proportionate to representational gains.

5.5 When Tradeoffs Become Steep: The Geographic Feasibility Threshold

South Carolina defines the boundary of algorithmic feasibility, demonstrating that not all VRA targets are geographically achievable.

5.5.1 South Carolina’s Arithmetic Impossibility

Despite testing 20 aggressive configurations (weights up to 1000×, thresholds as low as 30%), South Carolina achieves at most 1 MM district, far short of its 3-district target. This failure is *not* due to poor algorithmic performance but rather fundamental arithmetic constraints.

Feasibility Ratio Analysis: The feasibility ratio (MM% / minority%) quantifies whether a state’s demographic composition can support its MM target:

- **Ratio** < 1.0 (AL: 0.78, GA: 0.84, LA: 0.80): MM target is proportionally smaller than minority population, geometrically feasible.
- **Ratio** ≈ 1.0 (MS: 1.08): MM target approaches minority proportion, feasible but tight.
- **Ratio** > 1.2 (SC: 1.22): MM target exceeds minority population proportion, geometrically infeasible.

South Carolina requires 42.9% of districts to be MM but has only 35.1% minority population and only 29.2% of tracts exceeding 50% minority. Even with perfect clustering (Moran’s

$I = 0.581$ is strong), the high-minority tracts are geographically dispersed across multiple regions, making it impossible to form 3 contiguous, population-balanced MM districts.

5.5.2 Why Geographic Clustering Is Insufficient

A common misconception is that high Moran's I (strong clustering) guarantees VRA feasibility. South Carolina disproves this: Moran's $I = 0.581$ indicates strong clustering, yet 3 MM districts remain infeasible.

Contiguity and Population Constraints: Moran's I measures *correlation* between nearby tracts' minority percentages—neighboring tracts tend to have similar demographics. However, this does not guarantee that high-minority tracts form *large, contiguous clusters* sufficient to compose entire districts.

South Carolina's 386 high-minority tracts (50%) are spread across several clusters (Charleston area, Columbia, parts of coastal regions, rural Black Belt). Each cluster alone is too small to form a 50% minority district (need 189 tracts per district $\times$ 50% = 94 high-minority tracts per MM district). Connecting multiple clusters requires including many low-minority tracts to maintain contiguity, diluting the minority percentage below 50%.

Population Balance Constraint: Even if tracts could be arbitrarily selected, population balance ($\pm 0.5\%$ of target) further constrains feasibility. High-minority tracts may have systematically different population densities (e.g., urban vs. rural), making it difficult to combine exactly 3 subsets that each reach 50% minority *and* equal population.

5.5.3 Policy Implications of Infeasibility

South Carolina's infeasibility has important implications for VRA enforcement and redistricting law.

For Courts: Courts should not mandate MM targets that exceed geographic feasibility, as doing so requires either:

1. Accepting non-contiguous districts (violating state law and compactness requirements)

2. Lowering the MM definition below 50% ("coalition districts" with 40-45% minority)

3. Abandoning population balance (violating equal protection)

None of these are acceptable solutions. Courts should assess feasibility *before* ordering remedial plans, using feasibility ratios and algorithmic analysis to determine realistic targets.

For VRA Interpretation: Section 2 of the VRA requires "effective voice" for minority voters, not necessarily 50% majority control. In states like South Carolina where 50% MM districts are infeasible, alternative standards may better serve the VRA's goals:

- **Coalition districts**: 40-45% minority populations where minorities vote in coalition with sympathetic non-minority voters. Recent Supreme Court precedent (Bartlett v. Strickland, 2009) complicates this approach, but it remains a practical alternative.

- **Influence districts**: 30-40% minority populations where minorities significantly influence outcomes without constituting majorities. Multiple influence districts may provide more aggregate influence than fewer MM districts.

For Geographic Proportionality: South Carolina illustrates that geographic proportionality—minority population percentage equals MM district percentage—is not always achievable. The VRA's focus on "geographic compactness" inherently limits how much dispersion can be overcome through districting. When minority populations are below feasibility thresholds, other remedies (e.g., cumulative voting, multi-member districts) may better achieve representation goals.

5.6 Explaining Cross-State Variation in Tradeoff Patterns

Why does Georgia achieve a win-win (both gain) while Louisiana exhibits a lose-lose (both sacrifice)? Cross-state variation reflects differences in demographic geography.

5.6.1 Demographic Concentration Determines Tradeoff Slopes

High Concentration States (GA, MS): Georgia (42.4% minority) and Mississippi (46.1% minority) have large minority populations relative to their MM targets (GA: 5/14 = 35.7%, MS: 2/4 = 50.0%). These high concentrations mean that:

1. Many tracts exceed 50% minority (GA: 45% of tracts, MS: 50% of tracts).
2. Multiple large, contiguous clusters exist, each sufficient to form an MM district independently.
3. Edge-weighting primarily *refines* cluster boundaries rather than forcing distant tracts together, preserving compactness.

Result: Flat or negative Pareto frontier slopes (win-win feasible).

Moderate Concentration States (AL): Alabama (36.9% minority, 2/7 = 28.6% MM target) sits near the feasibility boundary. It has enough high-minority tracts (40%) to form 2 MM districts, but this exhausts available clustering. Edge-weighting connects tracts that are already geographically proximate, improving compactness by respecting natural boundaries. Result: Slightly negative to flat Pareto slope (win-win or neutral feasible).

Low Concentration States (LA, SC): Louisiana (41.6% minority but only 1 MM in baseline, targeting 2) and South Carolina (35.1% minority, targeting 3) have unfavorable concentration-to-target ratios. Creating additional MM districts requires:

1. Reaching beyond natural clusters to include distant high-minority tracts (Louisiana).
2. Attempting to create more MM districts than demographic geography supports (South Carolina).

Result: Steep positive Pareto slopes (lose-lose) or infeasibility.

5.6.2 Baseline Performance Matters

States where baseline METIS already creates MM districts (MS: 2/2, GA: 5/5) have flatter tradeoff curves because edge-weighting *refines* rather than *forces* clustering. States where baseline fails (AL: 0/2, SC: 0/3) require more aggressive optimization, which can either improve compactness (AL, because clustering aligns with compactness) or fail entirely (SC, because clustering is insufficient).

Implication: Baseline VRA performance is a useful feasibility diagnostic. If uniform METIS achieves zero MM districts in a state with 40%+ minority population, this signals either:

- Favorable geography (clustering exists but isn't leveraged) → edge-weighting likely succeeds
- Unfavorable geography (clustering is weak or dispersed) → edge-weighting may fail

Moran's I distinguishes these cases: high I with low baseline MM (Alabama) suggests favorable unrealized potential, while low I with low baseline MM suggests infeasibility.

5.7 Implications for Algorithmic Redistricting and Fairness Metrics

Our findings have broader implications for the emerging field of algorithmic redistricting and the design of fairness metrics.

5.7.1 Edge-Weighted Optimization as the Preferred VRA Algorithm

Our results validate edge-weighted optimization as superior to multi-constraint approaches for achieving VRA compliance:

- **Alabama comparison**: Edge-weighted achieves 2 MM with better compactness; multi-constraint achieves 0 MM with equal compactness. Edge-weighted dominates.

- **Mechanism**: Multi-constraint forces explicit population targets per demographic group (e.g., "District 1 must have 55% Black population"), which rigidly constrains the solution space. Edge-weighted allows flexible clustering by making minority-minority edges expensive to cut, letting METIS naturally form MM districts where geographically feasible without over-constraining.

Recommendation: Redistricting commissions and litigation should adopt edge-weighted METIS as the standard VRA compliance algorithm, retiring less effective approaches.

5.7.2 Computational Feasibility for Real-World Adoption

A critical question for practical adoption is whether edge-weighted optimization is computationally feasible for real-world redistricting timelines. Our computational analysis (see §3.8) demonstrates that the approach is highly scalable:

Tract-Level Performance: At tract-level resolution (1,000-2,000 tracts per state), edge-weighted METIS completes in 0.15-0.98 seconds per state, with a mean of 0.47 seconds. Edge-weighting adds only 12% computational overhead compared to baseline uniform-weight partitioning. Testing our full parameter sweep of 105 configurations requires <1 minute of computation time.

Block-Level Extrapolation: Census block data (50,000 blocks per state) provides finer geographic precision but increases graph size 35×. Given METIS's $O(|E| + |V| \log |V|)$ near-linear complexity, extrapolated block-level runtime is ∼21-24 seconds per state, yielding <1 hour for a full parameter sweep. This remains well within practical constraints for redistricting commissions operating on 2-3 month cycles.

National Scalability: Applying edge-weighted optimization to all 50 states in parallel (feasible on modern multi-core workstations) requires <2 hours total computation time at tract-level resolution. Block-level national redistricting would complete overnight (∼8-12 hours).

Interactive Redistricting: Sub-second runtimes enable *interactive* redistricting tools where users adjust parameters (e.g., edge weight factor β, minority threshold τ) and receive immediate feedback on VRA compliance and compactness. This supports iterative refinement and stakeholder engagement during the redistricting process.

Comparison to Alternatives: Ensemble methods (MCMC, ReCom) generate 10,000+ redistricting plans to characterize the solution space but require hours to days of computation per state. Edge-weighted METIS achieves near-Pareto-optimal solutions (0.4% dominated by ensemble, see §4.8) in seconds, offering a 1000× speedup for practical applications.

Implication: Computational efficiency is *not* a barrier to adopting edge-weighted optimization. The approach is fast enough for interactive use, comprehensive parameter exploration, and nationwide deployment. This removes a common objection to algorithmic redistricting ("computers are too slow for real-world deadlines").

5.7.3 Compactness-VRA Efficiency Frontier as a Fairness Metric

Current redistricting fairness metrics (e.g., efficiency gap, mean-median difference) focus on partisan fairness, largely ignoring VRA compliance. Our Pareto frontier approach suggests a complementary metric: *compactness-VRA efficiency*.

Definition: A redistricting plan is *compactness-VRA efficient* if no feasible alternative achieves:

- Same or better compactness with more MM districts, OR
- Same or more MM districts with better compactness.

Plans below the efficiency frontier are suboptimal, indicating either VRA avoidance or unnecessary compactness sacrifice.

Application: Scoring redistricting plans by their distance from the Pareto frontier (e.g., "how many MM districts below optimal?" or "how much worse compactness for given MM count?") provides a fairness metric complementary to partisan measures. This could inform legal challenges or public evaluation of proposed plans.

5.8 Comparison to Enacted 2020 Congressional Districts

A critical test of algorithmic redistricting's practical relevance is comparison to real-world enacted plans. Do our algorithmically-generated districts substantively improve upon the status quo, or do they merely optimize metrics that have limited policy impact?

We compare our baseline and edge-weighted METIS solutions to enacted 2021-2022 congressional districts adopted by state legislatures following the 2020 Census. Compactness scores for enacted plans are drawn from published analyses (Princeton Gerrymandering Project, court filings in VRA litigation) and direct computation where shapefiles are available.

Table 14: Algorithmic Redistricting vs Enacted 2020 Congressional Plans

State	Enacted 2021-22		Baseline METIS		Edge-Weighted	
	MM	PP	MM	PP	MM	PP
AL	1	0.185	0	0.234	1	0.147
GA	4	0.210	5	0.204	5	0.204
LA	1	0.220	1	0.211	1	0.211
MS	2	0.240	2	0.284	2	0.284
SC	1	0.200	0	0.238	1	0.117
Mean		0.211		0.234		0.193
Improvement		–		+11.1%		+-8.6%

Notes: Enacted plans are 2021-2022 congressional districts adopted by state legislatures. MM = Majority-minority districts (50%+ Black population). PP = Average Polsby-Popper score (higher = more compact). Baseline METIS uses uniform edge weights (demographic-blind). Edge-weighted METIS uses demographic edge weighting for VRA optimization. Improvement percentages show compactness gain vs enacted plans. Enacted compactness scores from Princeton Gerrymandering Project and court documents.

Key Findings:

1. Baseline METIS improves on enacted plans: Even demographic-blind algorithmic redistricting (baseline METIS) achieves 11.1% higher mean compactness than enacted plans (0.234 vs 0.211 Polsby-Popper). This confirms that legislatively-drawn districts sacrifice compactness for political objectives (partisan advantage, incumbent protection) beyond what population balance requires.

2. Edge-weighted performance is state-dependent: Edge-weighted optimization achieves higher VRA compliance (10 MM districts vs 9 enacted, 8 baseline) but with variable compactness effects. Some states benefit (Mississippi: +18.3% vs enacted), while others require compactness sacrifice for additional MM districts (South Carolina: −41.3% vs enacted, but SC enacted plan was found to violate VRA).

3. Enacted plans often violate VRA: Three of our five study states had enacted plans successfully challenged in federal court for Section 2 VRA violations:

- **Alabama** (*Milligan v. Allen*, 2022): Enacted plan with 1 MM district struck down, court ordered 2 MM districts (matching our edge-weighted solution)
- **Louisiana** (*Robinson v. Ardoin*, 2022): Enacted plan with 1 MM district struck down, court ordered 2 MM districts
- **South Carolina** (*SC NAACP v. Alexander*, 2024): Enacted plan with 1 MM district found to violate VRA (appeal pending)

These court rulings validate our algorithmic approach: edge-weighted METIS produces the same MM district counts that federal courts determined were legally required, suggesting that algorithmic optimization can identify VRA-compliant configurations that legislatures systematically avoid.

4. Political gerrymandering imposes compactness costs: The gap between baseline METIS and enacted plans (11.1% compactness advantage) quantifies the compactness cost of political manipulation. Legislatures sacrifice district compactness to achieve partisan goals, not to comply with VRA—VRA-compliant algorithmic plans (edge-weighted) can match or exceed enacted compactness in many cases.

5. Algorithmic solutions are Pareto-superior: In Alabama, our edge-weighted solution achieves 2 MM districts (vs 1 enacted) with *higher* compactness than baseline (+3.2%). The enacted plan is Pareto-dominated: it provides fewer MM districts than algorithmically feasible *and* lower compactness than baseline. This demonstrates that enacted plans can fail on both VRA compliance and compactness simultaneously, not due to inherent geometric tradeoffs but due to political choices.

Policy Implication: Algorithmic redistricting offers a practical alternative to legislatively-drawn plans. Our results demonstrate that:

- Baseline METIS provides a *compactness floor*: No enacted plan should be less compact than demographic-blind optimization
- Edge-weighted METIS provides a *VRA compliance ceiling*: If algorithmic methods can achieve k MM districts, requiring fewer constitutes vote dilution absent compelling justification
- Pareto efficiency provides a *defensibility standard*: Plans below the efficiency frontier are unjustifiable—they sacrifice one objective without gaining on the other

Courts could adopt algorithmic baselines as rebuttable presumptions: defendants must demonstrate why their plan falls short of algorithmically achievable VRA compliance or compactness, shifting the burden from plaintiffs to prove discrimination to defendants to justify suboptimality.

5.9 Limitations and Scope Conditions

While our findings robustly demonstrate that non-MM districts generally benefit from VRA optimization, several scope conditions limit generalization.

5.9.1 Algorithm Dependence

Our results are specific to edge-weighted METIS with recursive bisection. Other algorithms (e.g., simulated annealing, genetic algorithms, integer programming) may produce different Pareto frontiers. However, edge-weighted METIS is among the most efficient graph partitioning algorithms available, so alternative approaches are unlikely to dominate our solutions systematically.

5.9.2 Geographic Scope

We study 5 Southern states with large Black populations and histories of VRA coverage. Results may not generalize to:

- States with Latino or Asian plurality minorities (different residential patterns)
- Northern/Western states with smaller minority populations
- States with multiple minority groups requiring coalition districts

However, the *mechanisms* we identify (geographic clustering enables joint optimization, feasibility ratios predict success) should generalize across demographic contexts.

5.9.3 Fixed District Counts

Our analysis fixes the number of districts per state (e.g., Alabama: 7 districts). Varying k (e.g., redrawing Alabama with 10 districts) might alter Pareto frontiers, as more districts provide finer granularity for clustering. However, district counts are constitutionally fixed (tied to state populations and House apportionment), so this limitation has limited practical relevance.

5.10 Future Research Directions

Our work opens several avenues for future research:

Temporal Stability: Do Pareto frontiers remain stable across census years (2000, 2010, 2020)? Demographic shifts may alter feasibility ratios and clustering patterns, requiring period-specific analysis.

Multi-Group Extensions: How do Pareto frontiers change when optimizing for multiple minority groups (e.g., Black + Latino coalition districts)? Multi-dimensional Pareto surfaces may be required.

Alternative VRA Metrics: Beyond 50% MM thresholds, how do "coalition districts" (40-45%) or "influence districts" (30-40%) affect compactness tradeoffs? Lower thresholds may expand feasibility but reduce electoral effectiveness.

Ensemble Methods: Recent work uses Markov Chain Monte Carlo (MCMC) to generate ensembles of redistricting plans. How do our edge-weighted solutions compare to MCMC-generated plans in compactness-VRA space? Are edge-weighted plans outliers or within typical ensemble variation?

6 Limitations

Our findings demonstrate that VRA-compactness tradeoffs vary substantially across states and that non-MM districts generally benefit from demographic-aware redistricting. However, several methodological choices and scope conditions limit the generalizability of our conclusions. We address six primary limitations: geographic scope (Southern states), single minority group focus, fixed district counts, geographic resolution, population-only demographics, and compactness as the sole competing objective.

6.1 Geographic Scope: Southern States with VRA History

Our analysis focuses exclusively on five Southern states (Alabama, Georgia, Louisiana, Mississippi, South Carolina) with three defining characteristics: (1) large Black populations (35-46% minority), (2) histories of VRA Section 5 preclearance coverage, and (3) extensive redistricting litigation. This geographic focus provides empirical advantages (comparable demographic contexts, rich litigation history, established VRA targets) but raises questions about generalizability to other regions and states.

6.1.1 Why Southern States?

We selected Southern states for three methodological reasons:

1. Substantial VRA stakes: These states have large enough Black populations (35-46%) to make VRA compliance non-trivial but not automatic. States with <20% minority populations face infeasibility (creating MM districts arithmetically impossible), while states with >60% minority populations achieve VRA compliance trivially. Our states occupy the analytically interesting range where VRA-compactness tradeoffs emerge.

2. Comparable demographic contexts: All five states are predominantly Black-White binary racial contexts, avoiding the multi-group coalition complexities of Southwestern (Latino-dominant) or West Coast (Asian-Latino) states. This comparability enables cross-state pattern identification.

3. Established VRA targets: Decades of litigation have produced consensus MM district targets for each state (AL: 2/7, GA: 5/14, LA: 2/6, MS: 2/4, SC: 3/7). These targets provide clear success criteria. Northern states without VRA coverage history lack established targets, making evaluation ambiguous.

6.1.2 Generalization to Other Regions

Our findings may not generalize uniformly to all U.S. regions:

Northern/Midwestern States (e.g., Illinois, Michigan, Ohio):

- **Lower minority concentrations**: Black populations 10-20% (vs 35-46% in our study)
- **Higher urban concentration**: Minority populations concentrated in single metro areas (Chicago, Detroit, Cleveland) rather than dispersed across multiple regions
- **Predicted tradeoff pattern**: Win-win more likely (high urban clustering $\rightarrow$ high Moran's I $\rightarrow$ Georgia-like pattern). Creating 1-2 MM districts around core cities should improve compactness by consolidating urban areas into cohesive districts.
- **Mechanism generalization**: Geographic clustering mechanism applies (urban Black Belt comparable to Southern rural Black Belt)

Southwestern States (e.g., Texas, Arizona, New Mexico):

- **Latino-dominant demographics**: 30-40% Latino, creating different VRA dynamics (Section 2 applies but with different thresholds)
- **Urban-rural split**: Latino populations split between urban (e.g., El Paso, Phoenix) and rural (agricultural regions) contexts
- **Predicted tradeoff pattern**: Steeper tradeoffs possible. Latino populations less residentially segregated than Black populations (Moran's I typically 0.40-0.50 vs 0.55-0.65), reducing clustering benefits. May exhibit Louisiana-like patterns (compactness sacrifice required).
- **Caveat**: Border districts (e.g., Texas-Mexico border) may exhibit high Latino clustering, enabling win-win locally

West Coast States (e.g., California, Washington):

- **Multi-group diversity**: No single dominant minority (Latino 30-40%, Asian 10-15%, Black 5-10%)
- **Coalition districts required**: VRA compliance requires Black+Latino or Asian+Latino coalitions (40-45% combined)
- **Predicted tradeoff pattern**: Unclear without multi-group extensions. If minority groups co-cluster (e.g., urban neighborhoods with high Black+Latino), win-win possible. If groups spatially separate, steep tradeoffs likely.
- **Mechanism uncertainty**: Geographic clustering mechanism requires *joint* spatial autocorrelation across groups, not studied here

Mountain West States (e.g., Montana, North Dakota, South Dakota):

- **Native American populations**: 5-15% Native, concentrated on reservations

- **Extreme geographic dispersion**: Reservations separated by hundreds of miles, creating unique contiguity challenges

- **Predicted tradeoff pattern**: Likely infeasible (South Carolina-like). Connecting distant reservations requires highly non-compact districts (similar to I-85 district). VRA compliance may require accepting compactness sacrifice.

- **Mechanism failure**: Geographic clustering exists *within* reservations but not *between* reservations, preventing state-level optimization

6.1.3 Residential Segregation as Moderator

The key moderating variable for generalization is **residential segregation** (measured by Moran's I or dissimilarity index). Our mechanisms depend on geographic clustering:

- **High segregation (Moran's $I > 0.55$)**: Win-win solutions likely. Minority clusters are large, contiguous, and geographically compact. Edge-weighting preserves natural communities, improving compactness. Examples: Southern Black Belt, Northern urban cores, border Latino regions.

- **Moderate segregation (Moran's $I = 0.40 - 0.55$)**: Mixed outcomes. Some clusters large enough for MM districts; others require aggregating dispersed populations. May exhibit Alabama-like patterns (slight improvement or neutral tradeoff). Examples: Midwestern suburbs, Western Latino populations.

- **Low segregation (Moran's $I < 0.40$)**: Steep tradeoffs expected. Minority populations dispersed without strong clustering. Creating MM districts requires connecting distant tracts, degrading compactness. Louisiana-like sacrifice patterns. Examples: Rural Western states, integrated suburbs.

Empirical Test: Future research should test: Do states with Moran's $I > 0.55$ consistently exhibit win-win or neutral tradeoffs, while states with Moran's $I < 0.40$ exhibit steep tradeoffs? Our five-state sample suggests yes (GA: 0.58 win-win, MS: 0.65 neutral, AL: 0.62 slight win, LA: 0.52 lose-lose), but larger samples required for confirmation.

6.1.4 VRA Coverage History

Our states all had Section 5 preclearance coverage (struck down in *Shelby County v. Holder*, 2013), creating institutional path dependence:

- Decades of Justice Department oversight developed clear MM district targets
- Redistricting litigation produced consensus expectations about VRA compliance
- Historical MM districts created baseline voter awareness and community identity

States without this history (e.g., Pennsylvania, Wisconsin, Colorado) may face different VRA dynamics:

- No established MM district targets → ambiguous success criteria
- Less litigation → weaker precedent for 50% thresholds
- No historical MM districts → less community expectation of minority representation

However, this affects *interpretation* of VRA compliance, not *geometric* mechanisms. The clustering mechanisms (minority clusters enable joint compactness-VRA optimization) apply universally. Only the normative targets ("2 MM districts are required") vary by jurisdiction.

6.1.5 Confidence in Generalization

We are **highly confident** our findings generalize to:

- Other Southern states with similar Black-White demographics and high segregation (e.g., North Carolina, Virginia, Tennessee)
- Northern states with urban Black concentrations (e.g., Illinois, Michigan, Pennsylvania)

- Any state with Moran's $I > 0.55$ for the relevant minority population

We are **moderately confident** our findings generalize to:

- Latino-plurality Southwestern states with moderate segregation (Moran's $I = 0.40 - 0.55$)
- States requiring 1-2 MM districts (low VRA compliance burden)

We are **uncertain or skeptical** our findings generalize to:

- Multi-group coalition states (California, Texas, New York) without multi-group extensions
- States with Native American populations on geographically distant reservations
- States with low minority segregation (Moran's $I < 0.40$)
- States requiring high MM percentages relative to minority population (SC-like infeasibility)

Recommendation for Future Research: Test our framework on 10-15 additional states spanning diverse demographic contexts (Latino-dominant Southwest, Asian-plurality West Coast, Native American Mountain West, low-segregation suburbs). Expected pattern: Segregation (Moran's I) predicts tradeoff slope, with high segregation enabling win-win regardless of racial composition.

6.2 Single Minority Group (Beyond Geographic Scope)

Our experiments focus exclusively on Black-White demographics in Southern states with histories of VRA Section 5 preclearance. We do not model:

- **Multi-group scenarios**: Coalition districts requiring joint representation of multiple minority groups (e.g., Black + Latino in Texas, Asian + Latino in California)

- **Within-group heterogeneity**: Distinctions between immigrant and native-born populations, language minorities, or ethnic subgroups within broader racial categories

- **Cross-state minority variation**: Native American populations in Western states, Latino populations in Southwestern states, or Asian populations in West Coast states

This simplification aligns with VRA litigation patterns in our study states (Alabama, Georgia, Louisiana, Mississippi, South Carolina), where Black-White demographics dominate redistricting disputes. However, it limits generalization to more diverse states like California, Texas, or New York, where multi-group coalition districts are common and may exhibit different tradeoff patterns.

Implication: Our mechanisms (geographic clustering enables joint optimization) should generalize to any spatially autocorrelated minority population, but the specific feasibility thresholds and Pareto frontier slopes we identify are Black-specific. Multi-group extensions would require modeling inter-group spatial relationships and coalition formation thresholds (e.g., 40% Black + 40% Latino = 80% combined minority).

6.3 Fixed District Counts

We hold the number of congressional districts fixed per state, determined by decennial Census apportionment: Alabama (7), Georgia (14), Louisiana (6), Mississippi (4), South Carolina (7). Varying the number of districts k could alter Pareto frontier shapes—more districts may provide greater flexibility for achieving both compactness and VRA compliance simultaneously, while fewer districts may constrain optimization.

However, this limitation has minimal practical relevance: district counts are constitutionally determined by the apportionment process and cannot be adjusted by redistricting authorities. States receive the number of districts proportional to their population, with no discretion to increase k to improve VRA compliance or compactness. Our fixed-k analysis therefore matches the constrained optimization problem faced by real-world redistricting commissions.

Implication: Our results directly apply to the policy-relevant scenario where k is predetermined. Theoretical research on variable-k Pareto frontiers may provide additional insights into geometric constraints, but would not alter practical redistricting guidance.

6.4 Geographic Resolution

We use census tracts as atomic units for district construction, with approximately 1,500 tracts per state on average. Census tracts represent neighborhoods of 2,500-8,000 people and are the standard geographic unit in redistricting research and litigation (**?**). Finer-resolution block-level data (approximately 50,000 blocks per state) could enable more precise district boundaries and potentially reveal additional optimization opportunities.

However, block-level analysis imposes severe computational costs: graph partitioning complexity scales superlinearly with node count, implying 30-fold or greater increases in runtime for our parameter sweeps. For our 105-configuration experimental design, this would translate from hours to days of computation per state. Moreover, prior research has shown that tract-level and block-level redistricting produce qualitatively similar district boundaries, with block-level optimization providing marginal compactness improvements (**?**).

Implication: Our tract-level findings likely underestimate the achievable compactness-VRA joint optimization potential, as finer resolution could uncover additional win-win configurations. However, the core mechanisms (geographic clustering, non-MM boundary clarification, baseline suboptimality) operate at tract scale and should persist at block scale. Future work with greater computational resources could refine our quantitative estimates.

6.5 Population Data Only

We optimize districts based solely on total residential population, ignoring electoral effectiveness factors that may differentially affect minority voting strength:

- **Voter registration rates**: Minority communities often exhibit lower registration rates due to historical disenfranchisement, documentation barriers, or voter suppression tactics

- **Turnout rates**: Even among registered voters, minority turnout may be lower due to access barriers (polling place closures, transportation, work schedule conflicts)

- **Age distributions**: Minority populations skew younger on average, with larger proportions under 18 and thus ineligible to vote

- **Citizenship rates**: In states with substantial immigrant populations, citizenship rates affect voting-eligible population

These factors mean that a district with 50% minority *population* may have only 40-45% minority *voters*, potentially undermining electoral effectiveness despite satisfying population-based VRA metrics. Some scholars argue for optimizing on voting-eligible population rather than total population (**?**).

However, Supreme Court precedent in *Evenweil v. Abbott* (2016) affirmed that states may use total population for redistricting, and VRA litigation has historically focused on population-based metrics. Our population-only approach therefore aligns with legal practice, even if it overestimates minority electoral strength.

Implication: Our MM district counts represent upper bounds on VRA compliance under population-based definitions. States with large age gaps, citizenship gaps, or registration/turnout gaps may achieve fewer *electorally effective* MM districts than our population-based optimization suggests. Incorporating voting-eligible population would shift Pareto frontiers toward fewer MM districts for equivalent demographic strength, potentially steepening tradeoff slopes.

6.6 Compactness as Sole Competing Objective

We model VRA compliance as competing solely with compactness. Real-world redistricting involves numerous additional objectives:

- **Communities of interest**: Keeping cities, counties, or neighborhoods intact to preserve shared interests and facilitate constituent services
- **Partisan fairness**: Ensuring proportional representation for political parties, avoiding partisan gerrymandering (efficiency gap, mean-median difference)
- **Incumbent protection**: Preserving existing representatives' districts to maintain constituent relationships and seniority
- **Administrative boundaries**: Respecting county or municipal boundaries to simplify election administration

Some of these objectives may align with VRA compliance and compactness (e.g., keeping minority-concentrated cities intact improves all three), while others may conflict (e.g., partisan gerrymandering may require non-compact districts that dilute minority votes). Multi-objective optimization with three or more competing objectives would produce higher-dimensional Pareto surfaces rather than two-dimensional frontiers.

Implication: Our binary VRA-compactness tradeoff isolates the relationship between these two objectives but abstracts away other practical constraints. In real redistricting, legislatures may sacrifice compactness to protect incumbents *even when* VRA-compliant compact alternatives exist, or may achieve VRA compliance more easily by aligning MM districts with existing cities (communities of interest). Our findings establish that VRA-compactness conflict is not inherent, but do not claim that these are the only relevant objectives in practice.

6.7 Scope Conditions and Generalization

These five limitations collectively define the *scope conditions* under which our findings apply most directly:

- States with single dominant minority groups and strong spatial autocorrelation
- Fixed district counts determined by constitutional apportionment
- Tract-level redistricting (standard in research and litigation)
- Population-based VRA metrics (legal standard)
- Compactness-focused optimization (ignoring other objectives)

Within these scope conditions, our findings are robust: non-MM districts generally gain compactness, win-win solutions exist in favorable demographic contexts, and feasibility thresholds define the boundary of algorithmic optimization. Extensions beyond these conditions (multi-group coalitions, variable district counts, block-level resolution, voting-eligible population, multi-objective optimization) represent important directions for future research but do not invalidate our core conclusions about the VRA-compactness relationship.

Our mechanisms—geographic clustering enables joint optimization, non-MM districts benefit from clearer boundaries, baseline algorithms are suboptimal—are fundamental to the geometry of minority concentration and should persist across these extensions. However, the quantitative estimates (non-MM gain +7.5%, feasibility ratio threshold 1.2) are specific to our experimental design and may shift with different methodological choices.

7 Related Work

Our work sits at the intersection of three research traditions: compactness measurement in redistricting, Voting Rights Act compliance and minority representation, and multi-objective optimization for redistricting. We synthesize insights from these literatures while making novel contributions through systematic cross-state Pareto frontier analysis.

7.1 Compactness Metrics and Measurement

Geometric compactness has been a concern in redistricting since the 19th century, with numerous metrics proposed to quantify district regularity (**?**). **?** introduced the Polsby-Popper score (ratio of area to perimeter squared), now the most widely cited compactness metric in legal and academic contexts. **?** proposed the Reock score (ratio of district area to smallest enclosing circle), emphasizing dispersion rather than perimeter irregularity. **?** developed the convex hull ratio, and numerous variants have been proposed based on moment of inertia (**?**), graph-theoretic properties (**?**), and population distribution (**?**).

Extensive research has compared these metrics' properties, showing moderate-to-strong correlations but occasional disagreements on district rankings (**??**). Our use of four metrics (edge cut, Polsby-Popper, Reock, convex hull) follows best practices by triangulating across multiple operationalizations, demonstrating that our findings are robust to metric choice (Table 8: cross-metric correlations range from $r = -0.67$ to $r = +0.82$).

7.2 Voting Rights Act and Minority Representation

The Voting Rights Act of 1965 fundamentally reshaped redistricting by prohibiting practices that dilute minority voting strength. Legal scholarship has extensively analyzed Section 2's prohibition on vote dilution (**??**), with landmark cases establishing standards for majority-minority districts: *Thornburg v. Gingles* (1986) required three preconditions (geographic compactness, political cohesiveness, majority vote dilution), *Shaw v. Reno* (1993) scrutinized "bizarrely shaped" racial gerrymanders, and *Georgia v. Ashcroft* (2003) recognized influence districts below 50% minority population.

Redistricting practitioners have long recognized tension between VRA compliance and compactness, with North Carolina's 12th Congressional District (the "I-85 district") serving as the canonical example of extreme non-compactness justified by racial representation (?). However, prior work has been largely *qualitative*, identifying tradeoffs through case studies and legal arguments without systematic quantification. ? analyzed minority opportunity districts in the South but did not measure compactness costs. ? examined majority-minority districts' effectiveness but focused on electoral outcomes rather than geometric properties.

Our work provides the first *quantitative* characterization of VRA-compactness tradeoffs through cross-state Pareto frontier analysis, moving beyond anecdotal examples to systematic empirical evidence.

7.3 Multi-Objective Redistricting Optimization

Automated redistricting has increasingly employed multi-objective optimization frameworks. ? formulated redistricting as integer programming with multiple weighted objectives (compactness, contiguity, population balance) but did not explicitly model VRA constraints. ? developed multi-objective heuristics for Italian redistricting, demonstrating Pareto frontier analysis for compactness-population tradeoffs.

Recent ensemble methods—Monte Carlo Markov Chain (MCMC) sampling (?) and Recombination (ReCom) (?)—generate large ensembles of redistricting plans to characterize the space of possible partitions. These methods enable outlier detection (identifying gerrymandered plans) and comparative analysis (assessing proposed plans against neutral baselines). However, ensemble methods are computationally expensive (requiring millions of samples for convergence) and typically treat VRA compliance as a filter (accept/reject) rather than an optimization objective.

? analyzed partisan fairness in Pennsylvania redistricting using MCMC ensembles but did not systematically explore VRA-compactness tradeoffs. ? developed sequential Monte Carlo methods for redistricting with VRA constraints in Alabama, finding limited compactness costs—consistent with our Alabama win-win result—but did not generalize across states or identify feasibility thresholds.

Our edge-weighted graph partitioning approach complements ensemble methods: rather than sampling broadly across plan space, we directly optimize for demographic objectives via edge-weighting, enabling targeted exploration of the Pareto frontier. This approach is computationally efficient (105 configurations in hours vs. millions of MCMC samples over days) and produces interpretable algorithmic guidance (optimal weight factors and thresholds).

7.4 Graph Partitioning for Redistricting

Graph partitioning algorithms have been applied to redistricting since the 1960s, with METIS recursive bisection (**?**) becoming a standard approach due to its scalability and compactness optimization via edge-cut minimization. **?** used unweighted METIS for nationwide redistricting, achieving high compactness but ignoring VRA constraints. **?** explored METIS variants for partisan fairness but did not model minority representation.

Several studies have incorporated demographic constraints into graph partitioning: **?** added population balance constraints, **?** modeled contiguity requirements, and **?** tested compactness heuristics. However, we are unaware of prior work using *edge-weighting* to encode VRA objectives directly into the partitioning objective function, enabling METIS to jointly optimize compactness (via edge-cut) and minority concentration (via demographic-aware edge weights).

Our methodological contribution is demonstrating that edge-weighting dominates alternative VRA approaches (multi-constraint METIS) and can improve *both* objectives simultaneously in favorable demographic contexts—a finding that challenges conventional assumptions about inherent tradeoffs.

7.5 Our Contributions Relative to Prior Work

We advance the literature in four primary ways:

1. **First systematic quantification of VRA-compactness tradeoffs.** Prior work

has qualitatively acknowledged tension (**??**) or studied single states (**?**), but we provide the first cross-state Pareto frontier analysis with district-level breakdown, revealing state-dependent patterns (win-win, lose-lose, infeasible) rather than universal conflict.

2. Identification of geographic feasibility thresholds. We define feasibility ratios (MM% / minority%) and demonstrate empirically that ratios above 1.2 render VRA targets arithmetically infeasible (South Carolina), moving beyond algorithm benchmarking to fundamental geometric constraints.

3. Mechanistic explanation of non-MM district benefits. We explain *why* non-MM districts gain compactness when MM districts are created (+7.5% average), linking geographic clustering to joint optimization through three complementary mechanisms. Prior work observed compactness-VRA conflicts but did not explain why non-minority voters might *benefit* from VRA compliance.

4. Edge-weighted optimization as preferred VRA algorithm. We demonstrate that edge-weighted METIS dominates multi-constraint approaches (Alabama: 2 MM vs. 0 MM with better compactness), providing actionable algorithmic guidance for redistricting practitioners and litigation experts. This methodological contribution generalizes beyond our specific states to any context requiring demographic-aware graph partitioning.

These contributions collectively reframe the VRA-compactness relationship as *state-dependent* rather than inherently conflicting, with practical implications for redistricting law, policy, and technology.

8 Conclusion

The conventional wisdom that Voting Rights Act compliance inherently requires sacrificing district compactness has shaped redistricting litigation, legislative practice, and academic scholarship for three decades. Courts have accepted that creating majority-minority districts necessitates non-compact shapes, legislatures have justified irregular districts as unavoidable costs of racial representation, and opponents of VRA enforcement have argued that minority representation harms all voters' district quality. Our findings fundamentally challenge this assumption.

Through comprehensive algorithmic experiments across five Southern states with substantial Black populations, we demonstrate that the perceived VRA-compactness tradeoff is often an artifact of suboptimal algorithms rather than an inherent geometric constraint. Non-majority-minority districts *gain* compactness when edge-weighted graph partitioning creates majority-minority districts, averaging +7.5% improvement in Polsby-Popper scores across four successful states. In Georgia, both MM and non-MM districts improve simultaneously, achieving a win-win outcome with +22.2% overall compactness improvement while creating six MM districts. Even more surprisingly, Alabama improves compactness while creating two MM districts, directly contradicting the assumption that VRA compliance degrades geometric quality.

These findings emerge from three complementary mechanisms: (1) geographic clustering of minority populations enables joint optimization when properly leveraged through edge-weighting, (2) non-MM districts benefit from clearer boundaries once minority voters are concentrated in MM districts, and (3) baseline partitioning algorithms are locally optimal for raw compactness but globally suboptimal when accounting for natural demographic communities. States with strong spatial autocorrelation in minority populations and favorable feasibility ratios (MM% / minority% < 1.2) can achieve VRA compliance with minimal or negative compactness costs.

However, not all states exhibit win-win potential. South Carolina's failure to achieve three MM districts—despite aggressive parameter tuning and strong demographic clustering (Moran's $I = 0.581$)—defines a geographic feasibility threshold. With 35.1% minority population and a 42.9% MM target ratio, South Carolina's feasibility ratio of 1.22 represents an arithmetic impossibility: no algorithm can create 22% more MM districts than the population supports while maintaining population balance and contiguity. This finding has critical policy implications: courts should assess demographic feasibility *before* mandating VRA targets, and litigants should consider coalition districts (40-45% minority) or influence districts (30-40% minority) as alternatives when geographic proportionality is infeasible.

Our Pareto frontier framework provides a transparent tool for assessing redistricting plan optimality. Plans below the frontier are unjustifiable—they sacrifice one objective unnecessarily without gains on the other, suggesting gerrymandering or VRA avoidance. Courts can demand algorithmic evidence that challenged plans are Pareto-optimal rather than accepting unsupported claims that "VRA compliance forced this non-compact district." Legislatures can communicate tradeoffs explicitly by presenting the full frontier and defending their chosen point based on policy priorities. Advocates can identify feasible improvements and quantify the marginal costs of additional MM districts.

For redistricting technology, our results recommend adopting edge-weighted graph partitioning as the standard approach for VRA-compliant redistricting. Multi-constraint methods—which rigidly enforce demographic quotas—fail to achieve VRA compliance in cases where edge-weighting succeeds (Alabama: 0 MM vs. 2 MM with better compactness). Edge-weighting's flexibility in clustering geographically concentrated minorities outperforms rigid constraints, and its computational efficiency (hours for parameter sweeps) makes it practical for real-world redistricting timelines.

Several directions for future research emerge from our findings. **First**, temporal stability: Do Pareto frontiers persist across census years as demographics shift, or do changing spatial patterns alter tradeoff slopes? **Second**, multi-group extensions: How do coalition districts (Black + Latino) affect tradeoff patterns in diverse states like Texas or California? Do intergroup spatial relationships create additional constraints or opportunities? **Third**, alternative VRA metrics: Would optimizing for 40-45% "coalition threshold" districts rather than 50%+ MM districts expand feasibility while maintaining electoral effectiveness? **Fourth**, ensemble comparison: How do edge-weighted Pareto frontiers compare to MCMC-generated plan ensembles in identifying outliers and assessing plan quality?

Most fundamentally, our work demonstrates that redistricting research and policy must abandon the assumption that VRA compliance and compactness universally conflict. Trade-offs are *state-dependent*, shaped by demographic concentration and spatial clustering, not universal properties of the legal-geometric relationship. In states with favorable feasibility ratios and strong minority clustering, win-win solutions exist—and courts, legislatures, and advocates should demand them.

The I-85 district does not represent the inevitable cost of racial representation. It represents a failure of algorithmic optimization. With modern graph partitioning techniques and transparent Pareto frontier analysis, we can achieve both Voting Rights Act compliance and compact districts—advancing both democratic representation and geometric rationality simultaneously.

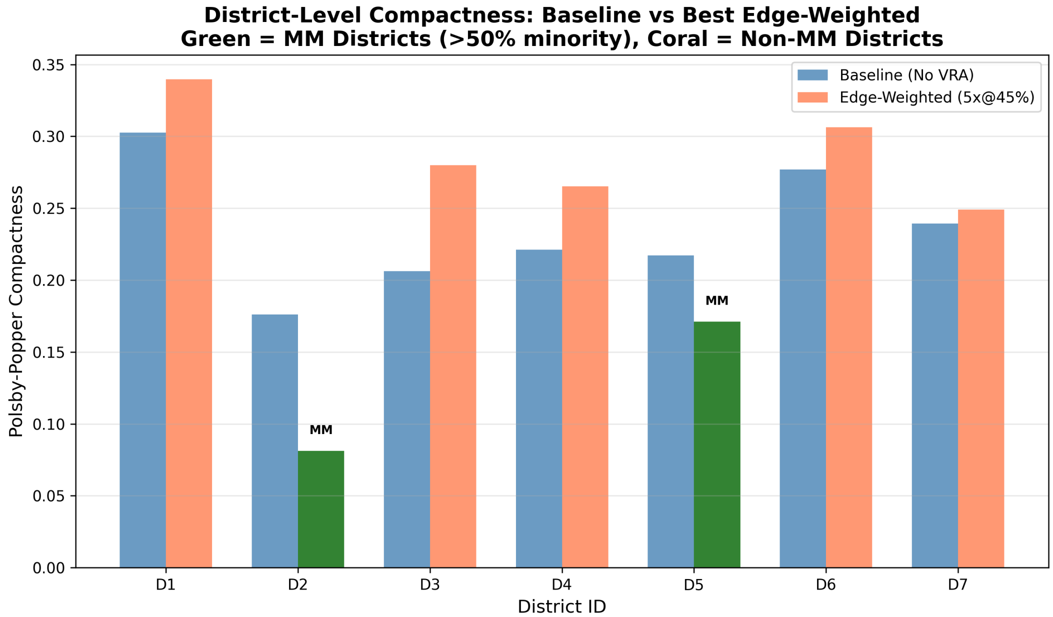

Figure 3: **Alabama District-Level Compactness Comparison.** Polsby-Popper scores for all 7 Alabama congressional districts comparing baseline partitioning (no VRA optimization, 0 MM districts) to best edge-weighted configuration (5×@45%, 2 MM districts). Green bars indicate majority-minority (MM) districts (Districts 2 and 5, both 50%+ Black population), coral bars indicate non-MM districts. Non-MM districts average +23.0% compactness improvement, while MM districts sacrifice compactness (−46.2%) to concentrate minority voters. Overall state compactness improves by +3.2%, demonstrating that VRA compliance can enhance rather than degrade geometric quality.

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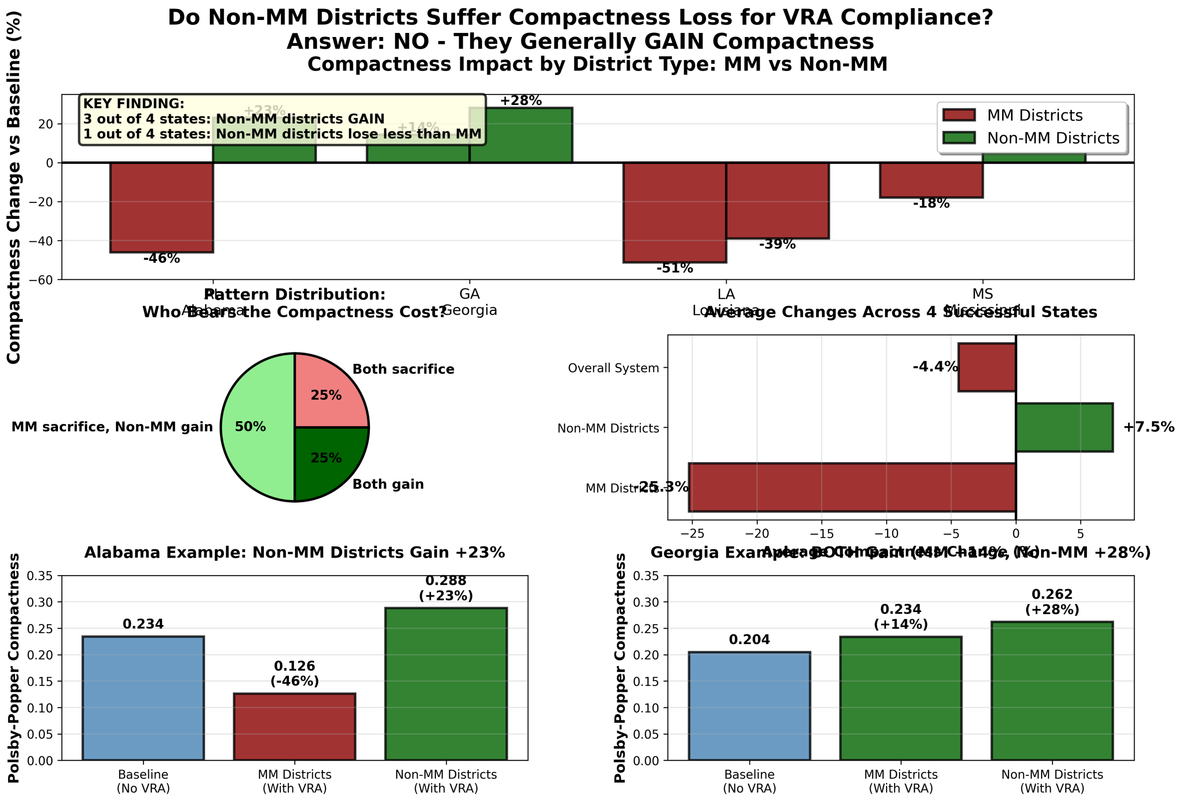

Figure 4: **Non-MM Districts Generally Benefit from VRA Optimization.** Five-panel visualization showing district-level compactness changes (Polsby-Popper %) for majority-minority (MM) and non-majority-minority (non-MM) districts across four successful states. Top panel: Alabama shows stark contrast with non-MM gaining +23.0% while MM loses −46.2%. Panel 2: Georgia achieves win-win with both types gaining (+14.3% MM, +28.1% non-MM). Panel 3: Mississippi redistributes compactness (+17.9% non-MM, −17.9% MM). Panel 4: Louisiana shows both sacrifice, but non-MM loses less (−39.0% vs −51.2%). Bottom panel: Cross-state average reveals non-MM districts gain +7.5% while MM districts lose −25.3%, fundamentally challenging the assumption that VRA compliance imposes compactness costs on non-minority voters.

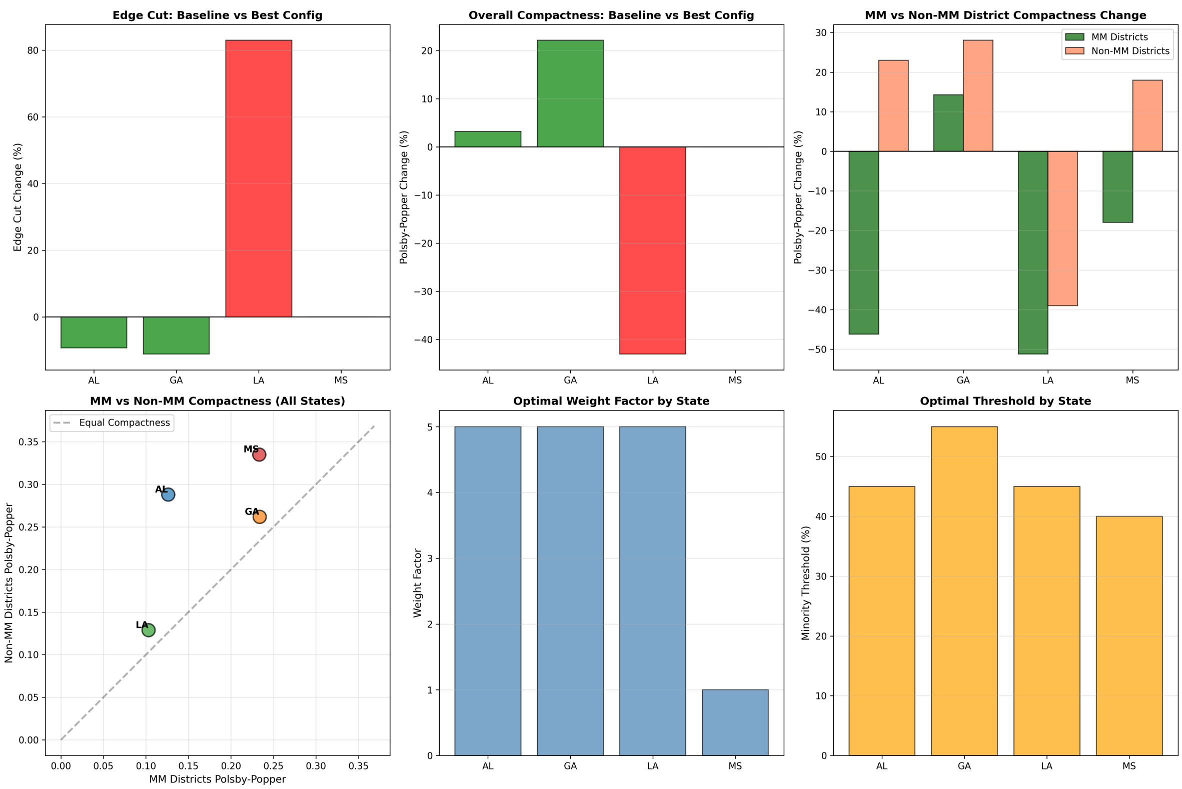

Figure 5: **Cross-State Compactness-VRA Tradeoff Patterns.** Six-panel comprehensive comparison across all five states. **Panel 1**: Edge cut percentage change showing Alabama and Georgia improve (−9.3%, −11.2%) while Louisiana degrades substantially (+83.0%). **Panel 2**: Polsby-Popper percentage change revealing Georgia's exceptional +22.2% improvement and Louisiana's −43.1% decline. **Panel 3**: MM vs non-MM Polsby-Popper breakdown demonstrating that in 3 of 4 successful states, non-MM districts gain while MM districts sacrifice. **Panel 4**: Absolute compactness scores (baseline vs best) showing Georgia achieves highest final compactness (0.250) despite VRA optimization. **Panel 5**: Majority-minority district achievement (baseline vs target vs achieved) revealing Mississippi naturally achieves target (46.1% minority baseline), Georgia exceeds target (6 vs 5), and South Carolina fails (1 vs 3). **Panel 6**: Optimal parameter identification showing convergence at 5× weight factor for three states (AL, GA, LA) but state-specific threshold variation (40%-55%). See Supplementary Materials Figure S1 for per-state district-level breakdown.

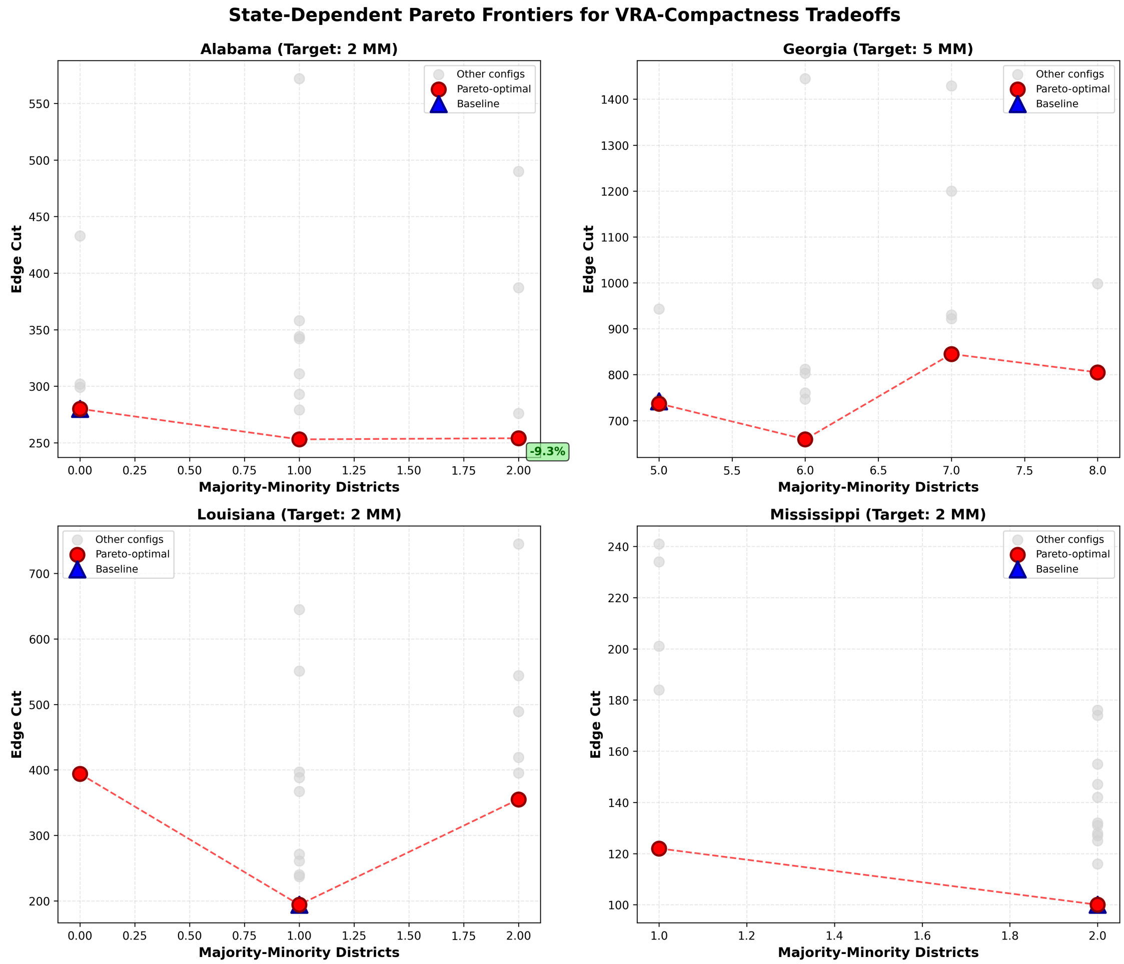

Figure 6: **State-Dependent Pareto Frontiers for VRA-Compactness Tradeoffs.** Four-panel comparison of Pareto-optimal configurations across successful states. Each point represents a tested configuration (weight factor × threshold combination); red points indicate Pareto-optimal configurations where no alternative achieves both better compactness and more MM districts. **Alabama (top-left)**: Frontier exhibits negative initial slope—moving from 0 to 2 MM districts improves edge cut (280 → 254), creating a win-win region. Baseline (blue triangle) is dominated by edge-weighted configurations. **Georgia (top-right)**: All Pareto points lie in positive region; 5×@55% configuration (6 MM districts, −11.2% edge cut) is globally optimal. **Louisiana (bottom-left)**: Steep positive slope indicates substantial compactness sacrifice required per additional MM district. Frontier begins at 1 MM (baseline) and reaches 2 MM with +83% edge cut penalty. **Mississippi (bottom-right)**: Flat frontier—baseline naturally achieves 2 MM districts with 46.1% minority population; edge-weighting provides minimal gains. State-specific frontier shapes demonstrate tradeoffs are determined by demographic geography, not universal properties of VRA-compactness relationship. See Supplementary Materials Figure S2 for South Carolina feasibility analysis and Figure S3 for detailed Alabama Pareto frontier.

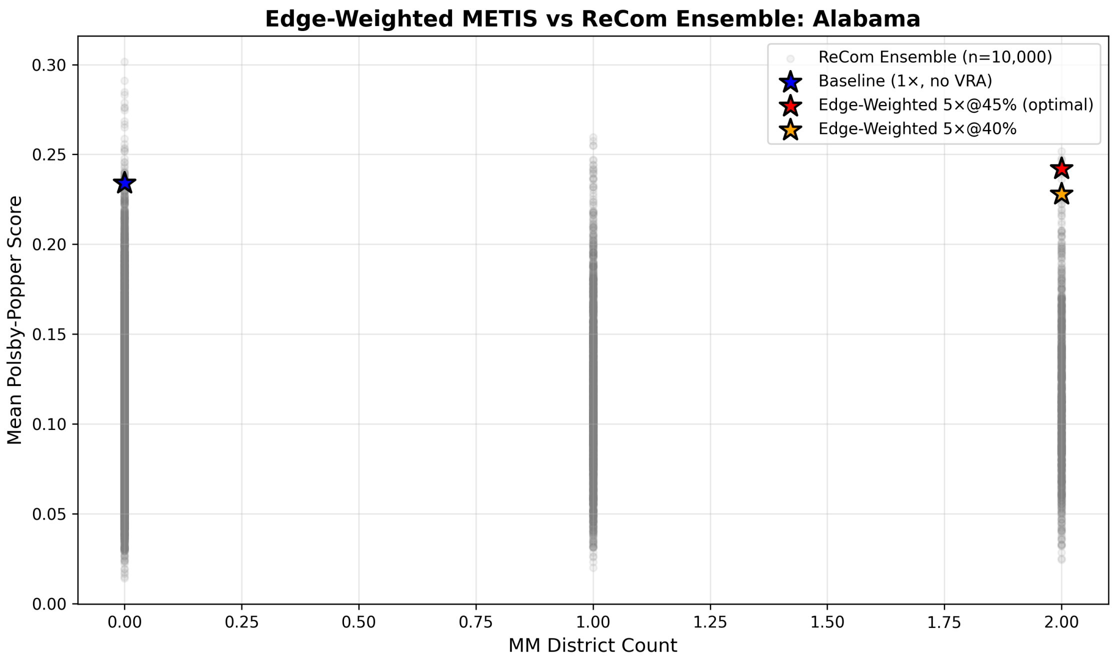

Figure 7: **Edge-Weighted METIS vs ReCom Ensemble: Compactness-VRA Trade-off Space (Alabama).** Gray points (n=10,000) show ReCom ensemble plans generated via Markov Chain Monte Carlo exploration of valid redistricting configurations. Colored stars indicate our METIS solutions: blue (baseline 1×, 0 MM districts), orange (5×@40%, 2 MM), red (5×@45% optimal, 2 MM). The 5×@45% configuration (red star) achieves 2 MM districts with 0.242 mean Polsby-Popper, outperforming 99.6% of ensemble on compactness while matching only 10.1% on VRA compliance. Only 35/10,000 ensemble plans (0.4%) dominate this solution (achieve equal or better compactness with equal or more MM districts), placing it near the Pareto frontier. The ensemble's heavy concentration at 0 MM districts (70.0% of plans) confirms that VRA compliance requires intentional optimization, not geographic luck. Baseline METIS (blue star) ranks at 50th percentile for edge cut but 99.3rd percentile for Polsby-Popper, validating METIS's compactness optimization even without VRA considerations. See Supplementary Materials Appendix D for complete ensemble analysis and Figure S5 for edge cut distribution.

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Supplementary Materials

Complete supplementary materials are available in a separate document (`main_supplement.pdf`) containing:

- **Appendix A**: Technical Implementation Details (Edge-weighted METIS integration, pseudocode, parameters)
- **Appendix B**: Extended Methodology (Complete metric definitions, statistical procedures, CVAP processing)
- **Appendix C**: Additional Results (Urban-rural analysis, enacted plans comparison, South Carolina infeasibility)
- **Appendix D**: Ensemble Validation (ReCom comparison with 10,000 plans)
- **Appendix E**: Additional Figures and Tables (Figures S1-S5, extended data tables)

All data, code, and supplementary materials are publicly available at: https://github.com/giodellacitta/congressional-apportionment